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Moreno et al.

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(54) **SENSOR DRIVEN SECURE DISPENSING UNIT**

(58) **Field of Classification Search**
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A61G 12/00 (2006.01)
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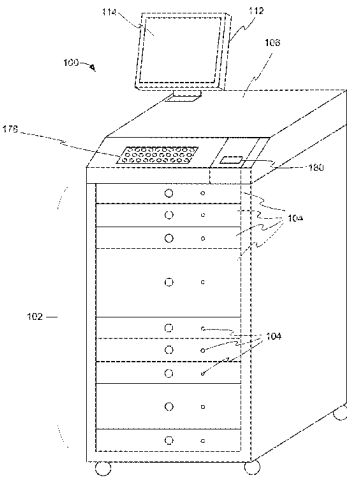
(52) **U.S. Cl.**
CPC **G16H 20/13** (2018.01); **A61G 12/001** (2013.01); **G16H 10/60** (2018.01); **G16H 40/20** (2018.01);

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(57) **ABSTRACT**

A secure dispensing unit includes a housing comprising a first end and a second end and a number of storage assemblies. The number of storage assemblies are arranged in parallel with one another within the housing. Each storage assembly includes a number of compartments arranged linearly along an axis of the housing with each of the compartments being configured to store an item. A cover is coupled with the housing and extends between the first end and the second end such that the cover is positioned over the compartments. An actuator is coupled with the cover, the actuator being configured to retract the cover to draw a distal end of the cover from the first end toward the second end to expose a selected number of compartments.

15 Claims, 17 Drawing Sheets



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H04N 19/13 (2014.01)
H04N 19/159 (2014.01)
H04N 19/174 (2014.01)
H04N 19/176 (2014.01)
H04N 19/426 (2014.01)
H04N 19/44 (2014.01)
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H04N 19/61 (2014.01)
H04N 19/625 (2014.01)
H04N 19/82 (2014.01)
H04N 19/96 (2014.01)

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H04N 19/82 (2014.11); *H04N 19/96* (2014.11)

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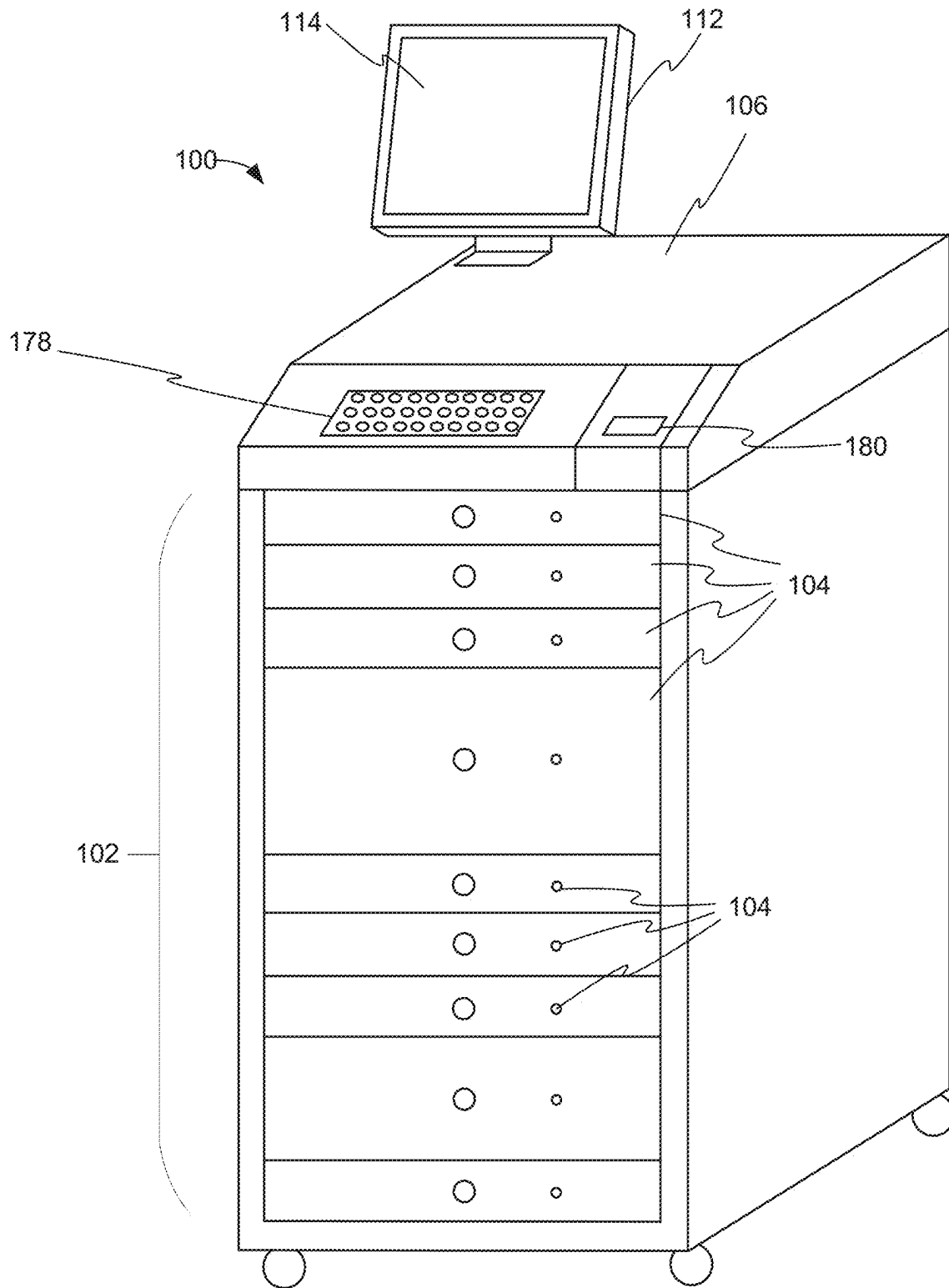
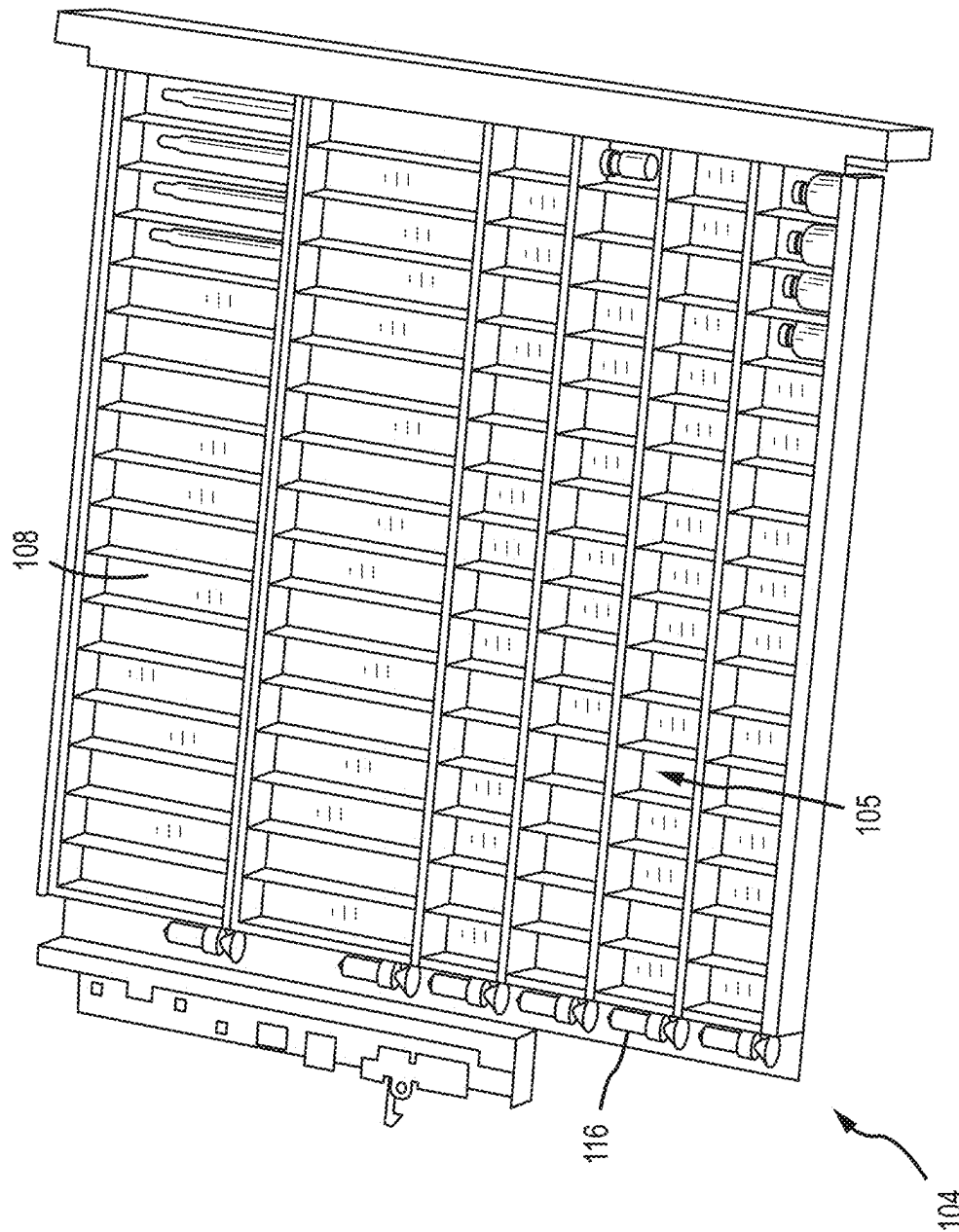
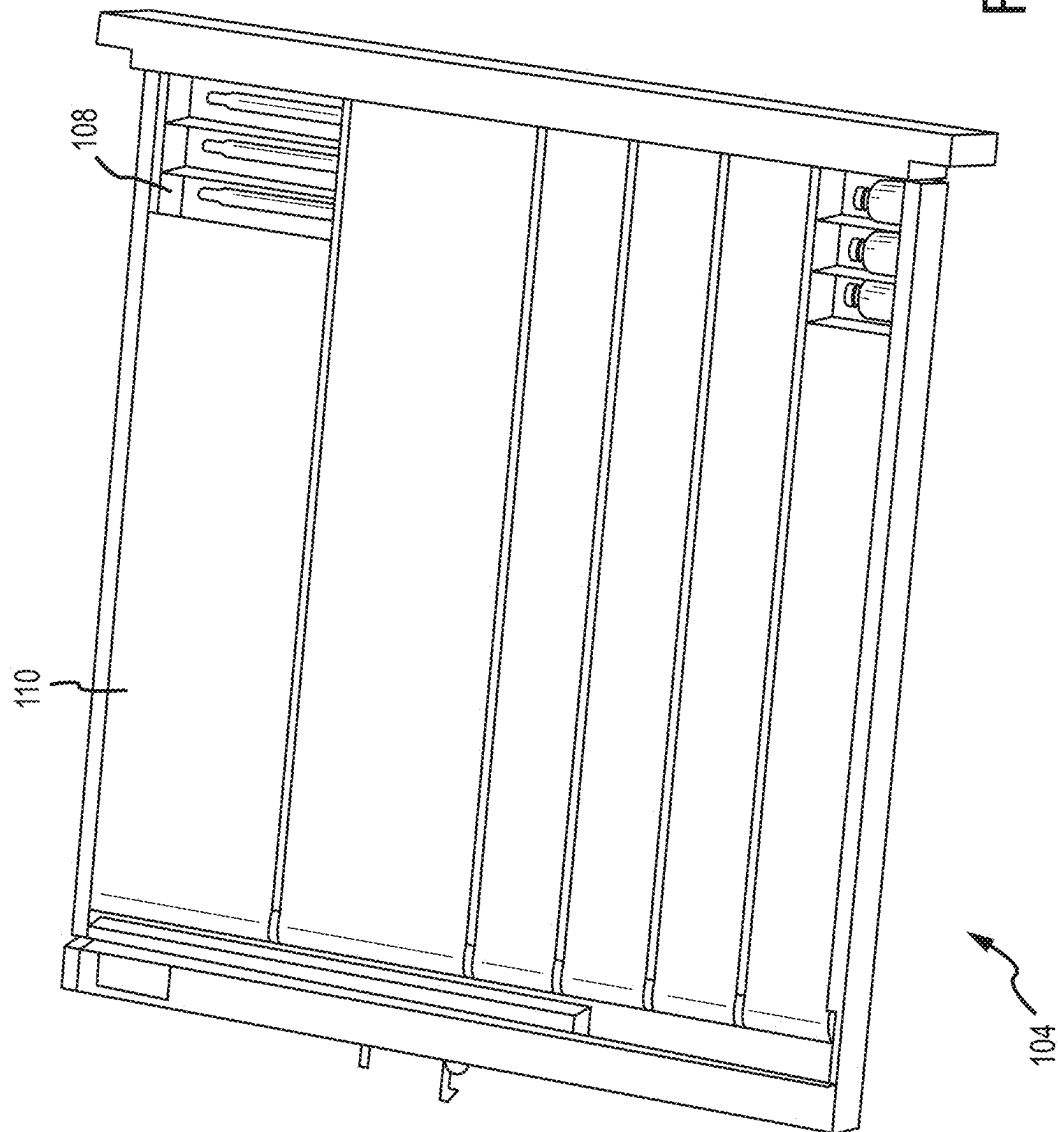


FIG. 1





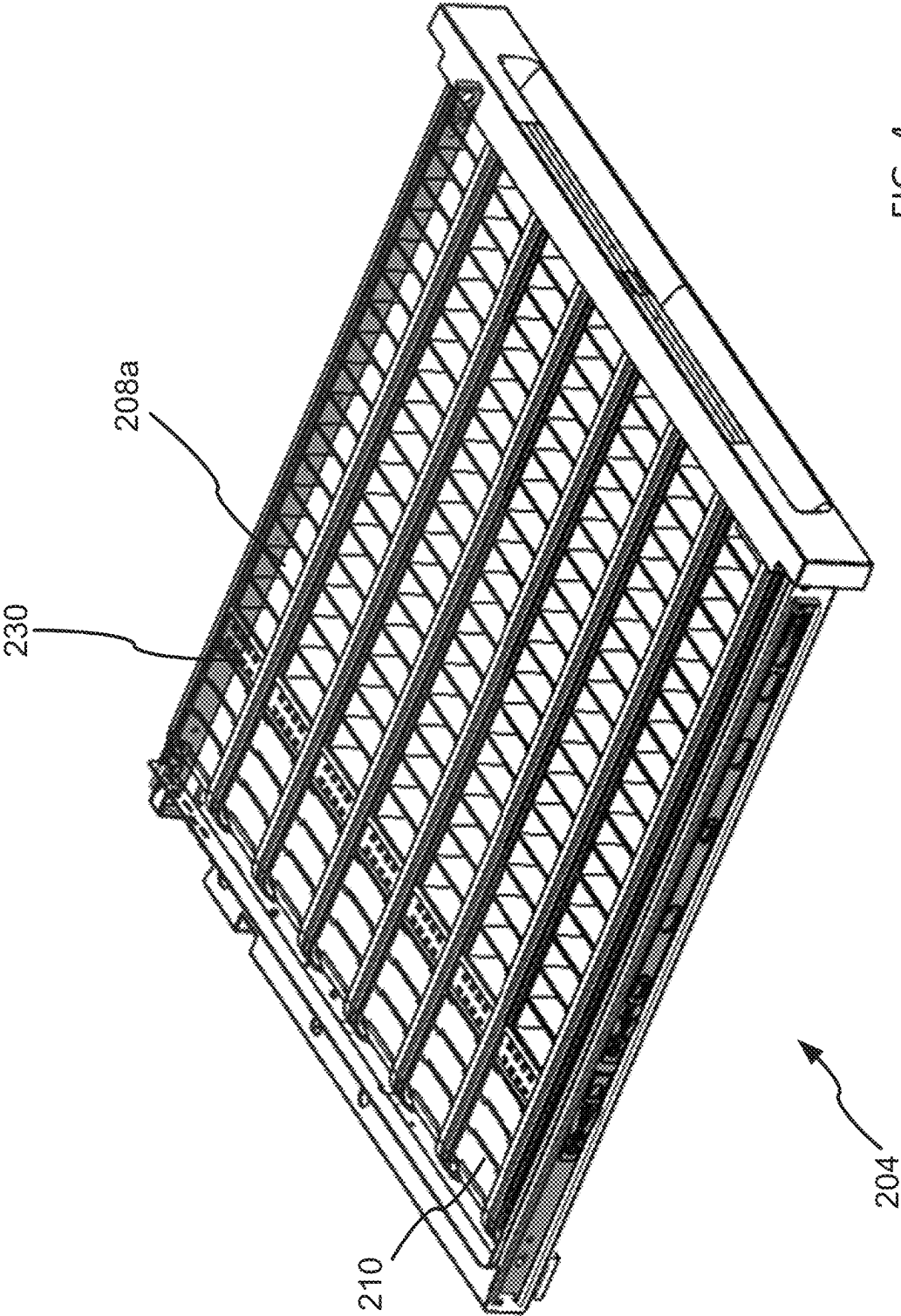


FIG. 4

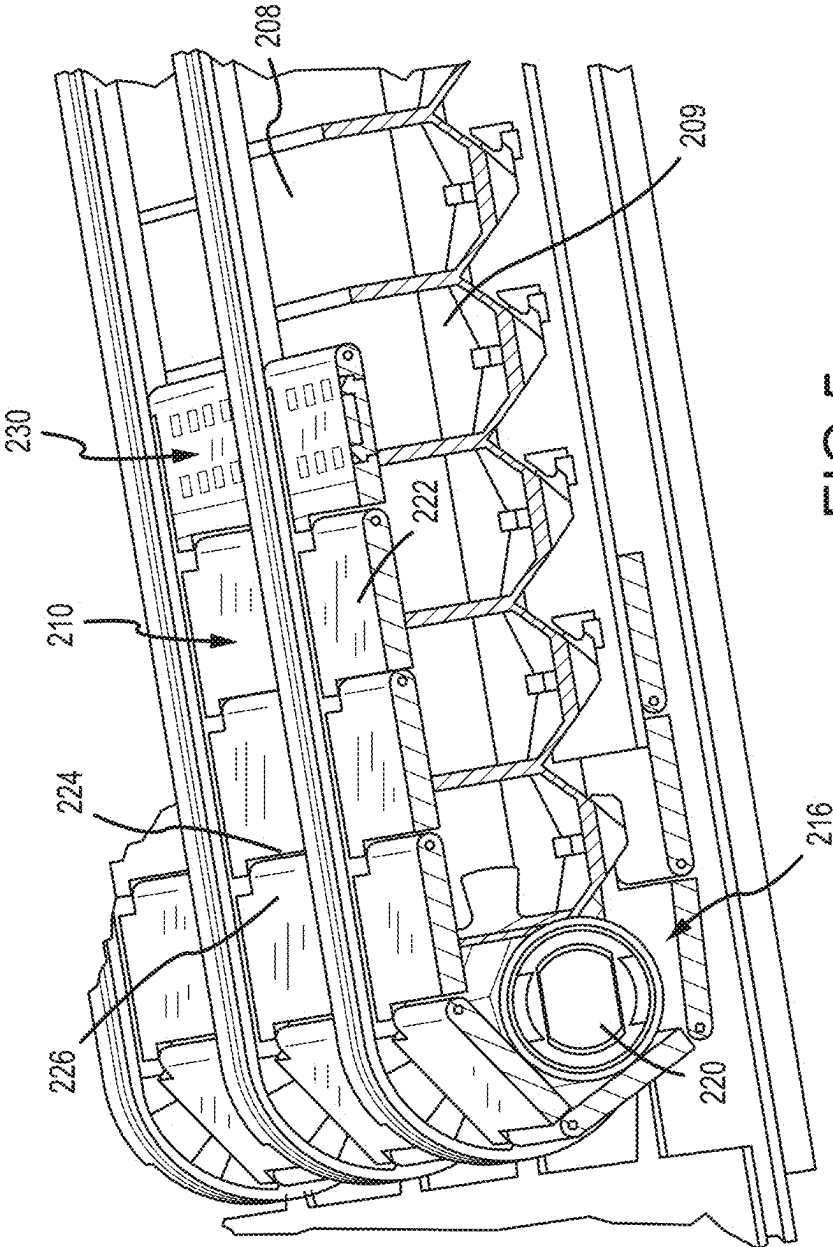


FIG. 5

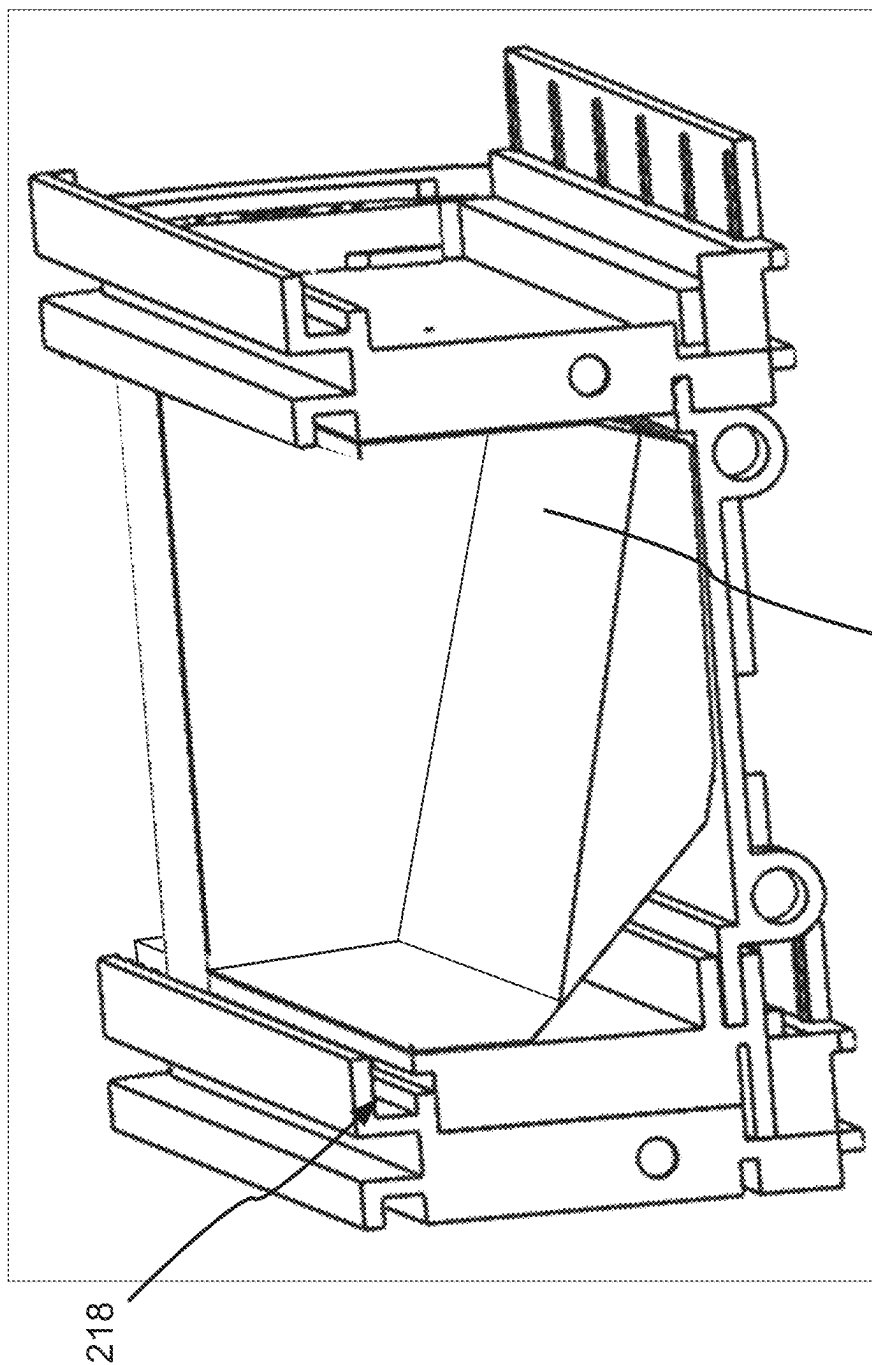


FIG. 6

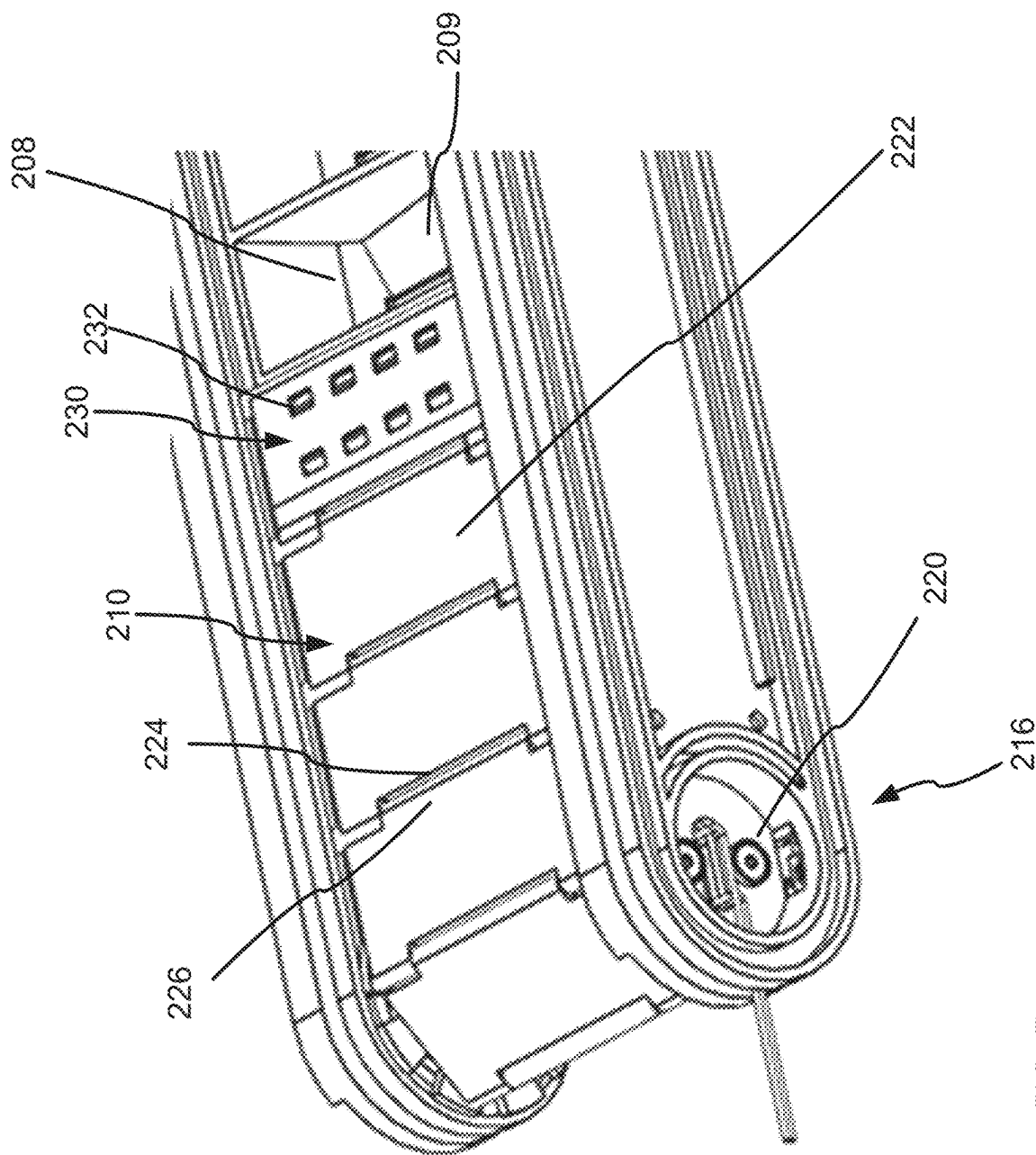


FIG. 7

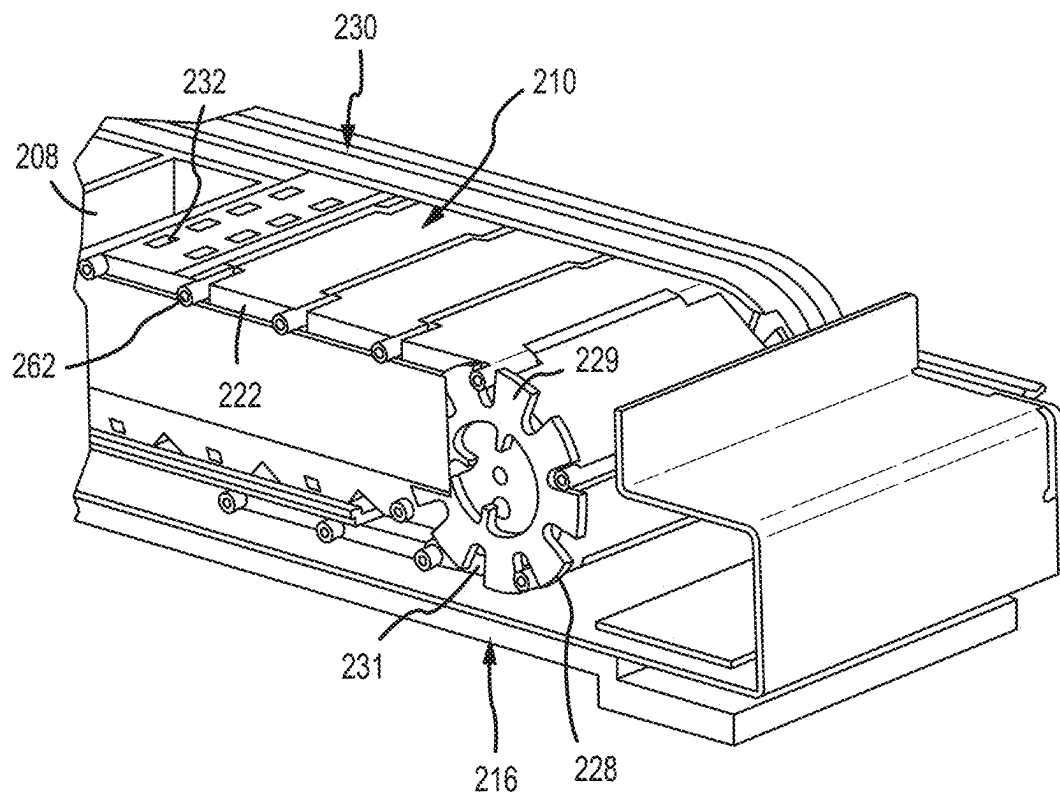


FIG.8

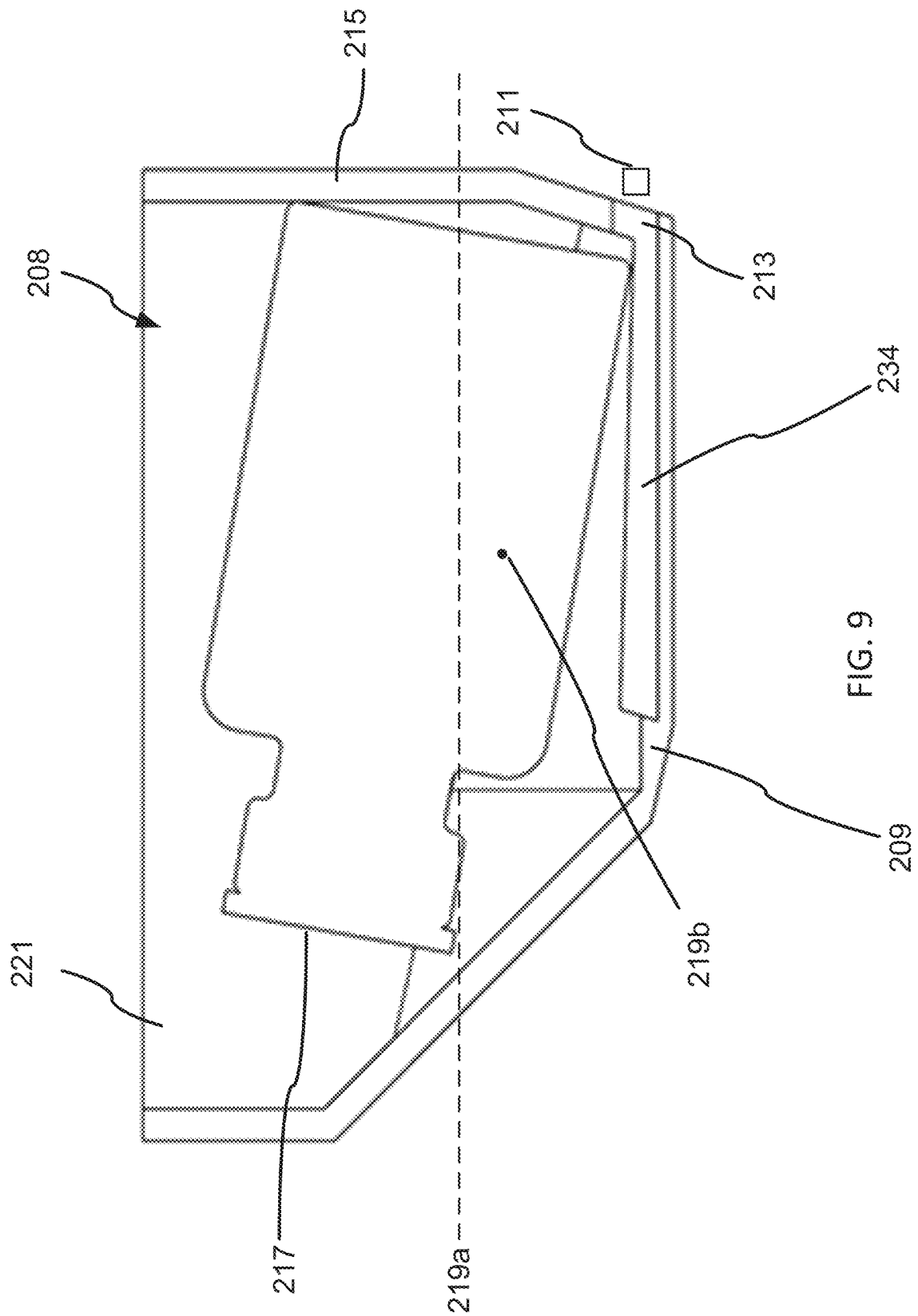


FIG.10A

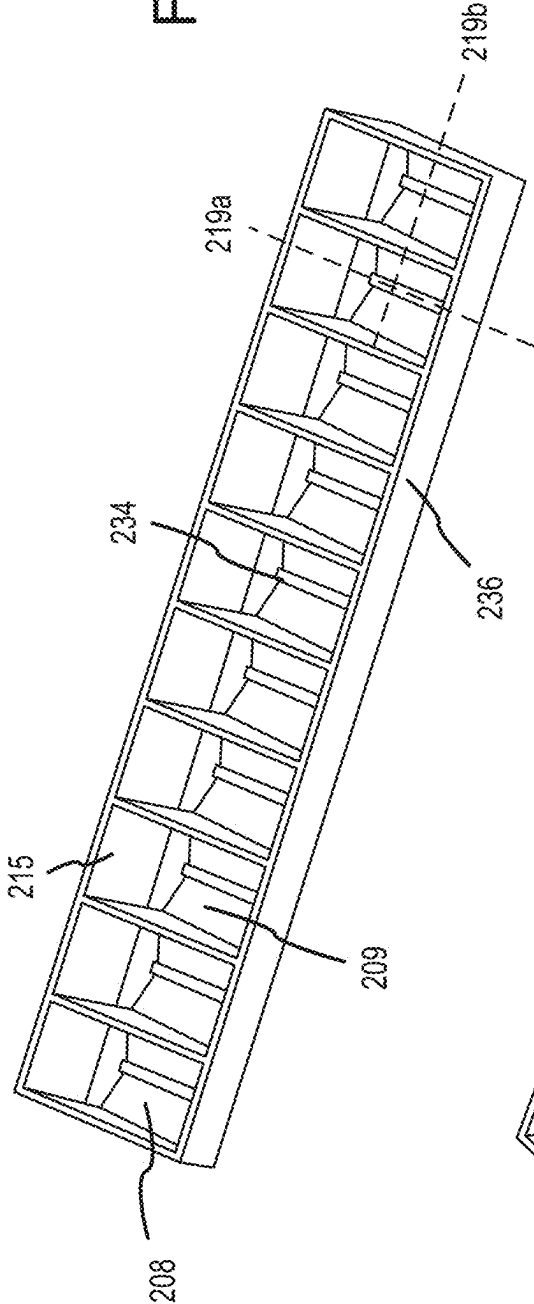
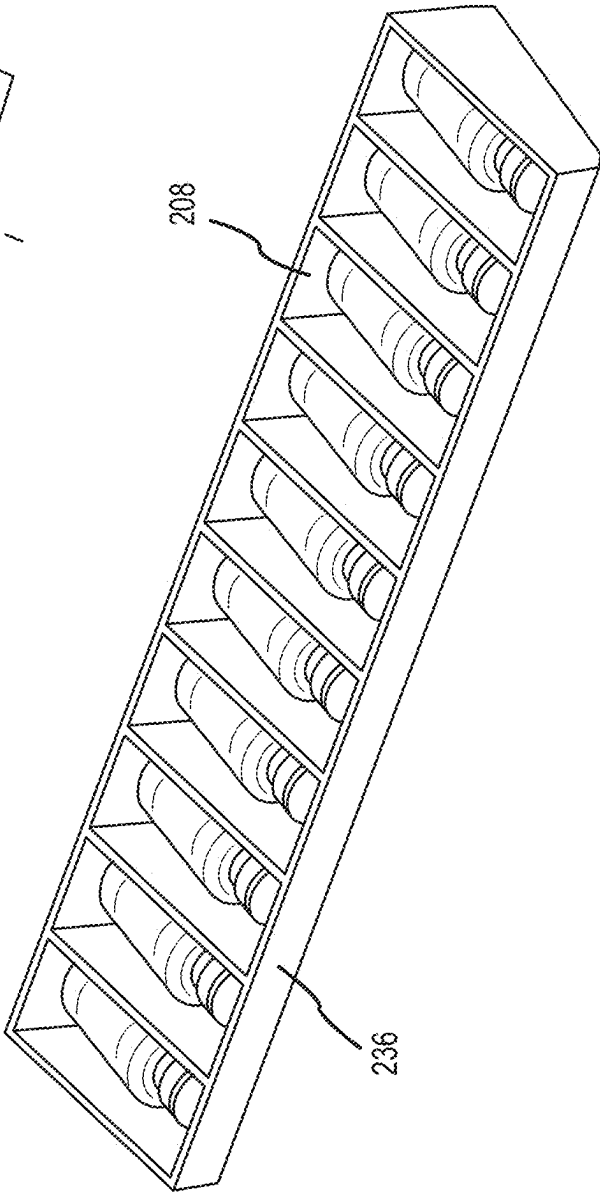


FIG.10B



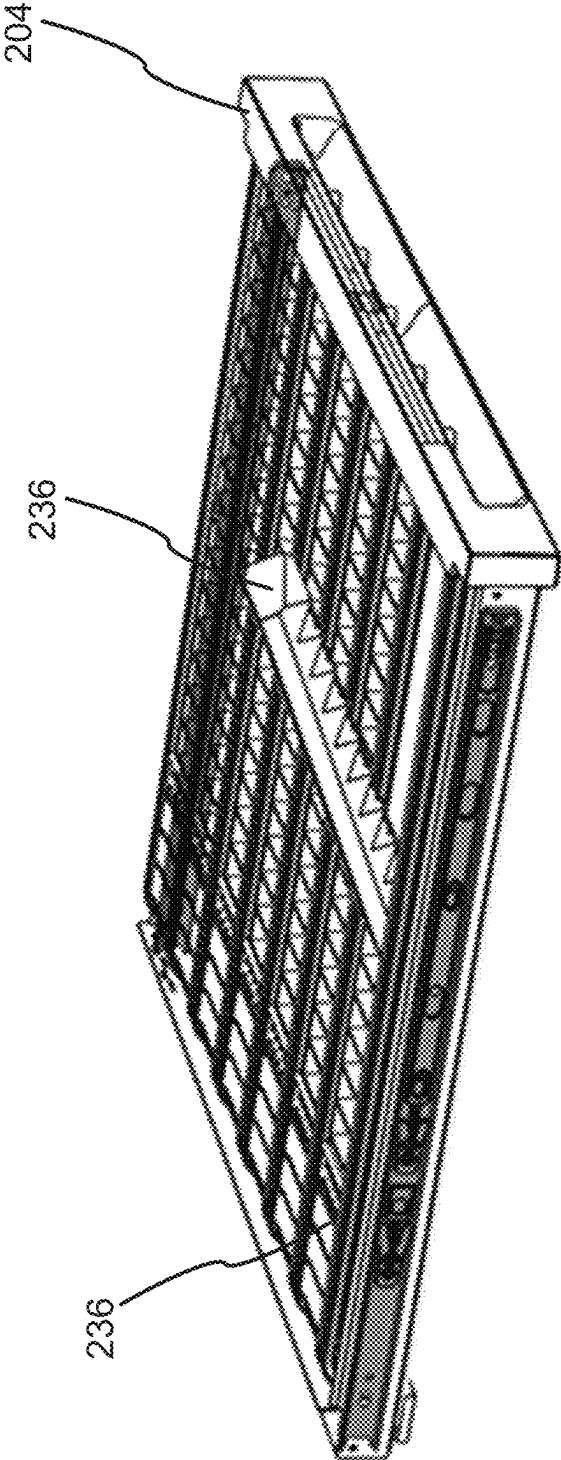


FIG. 11

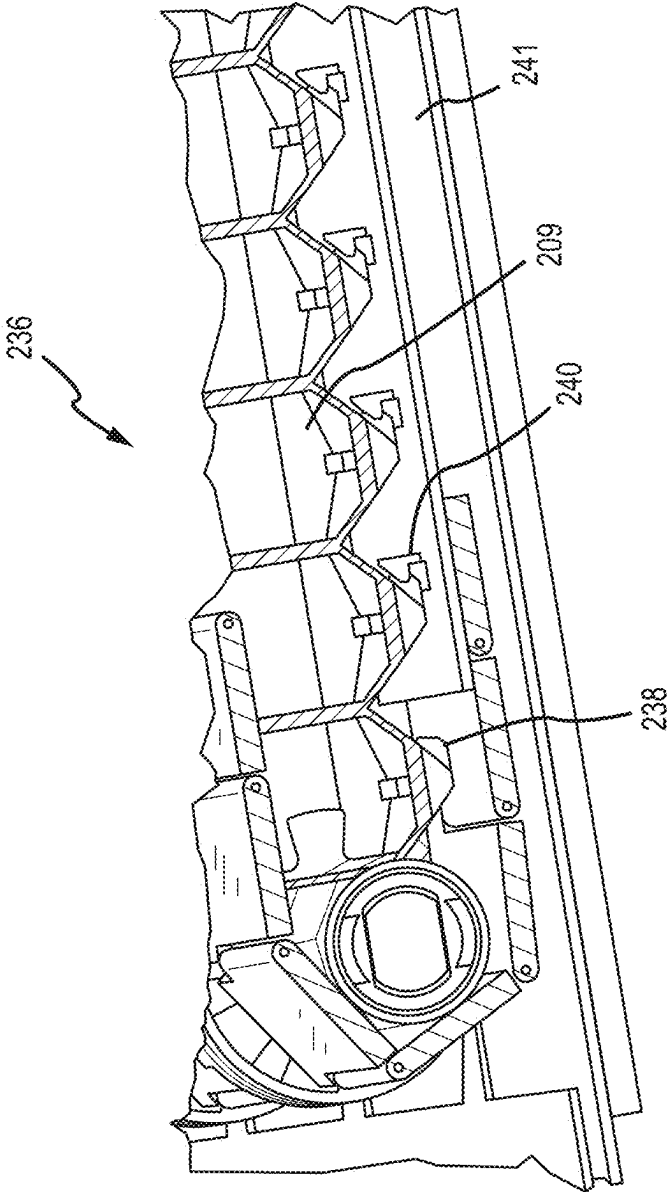


FIG.12

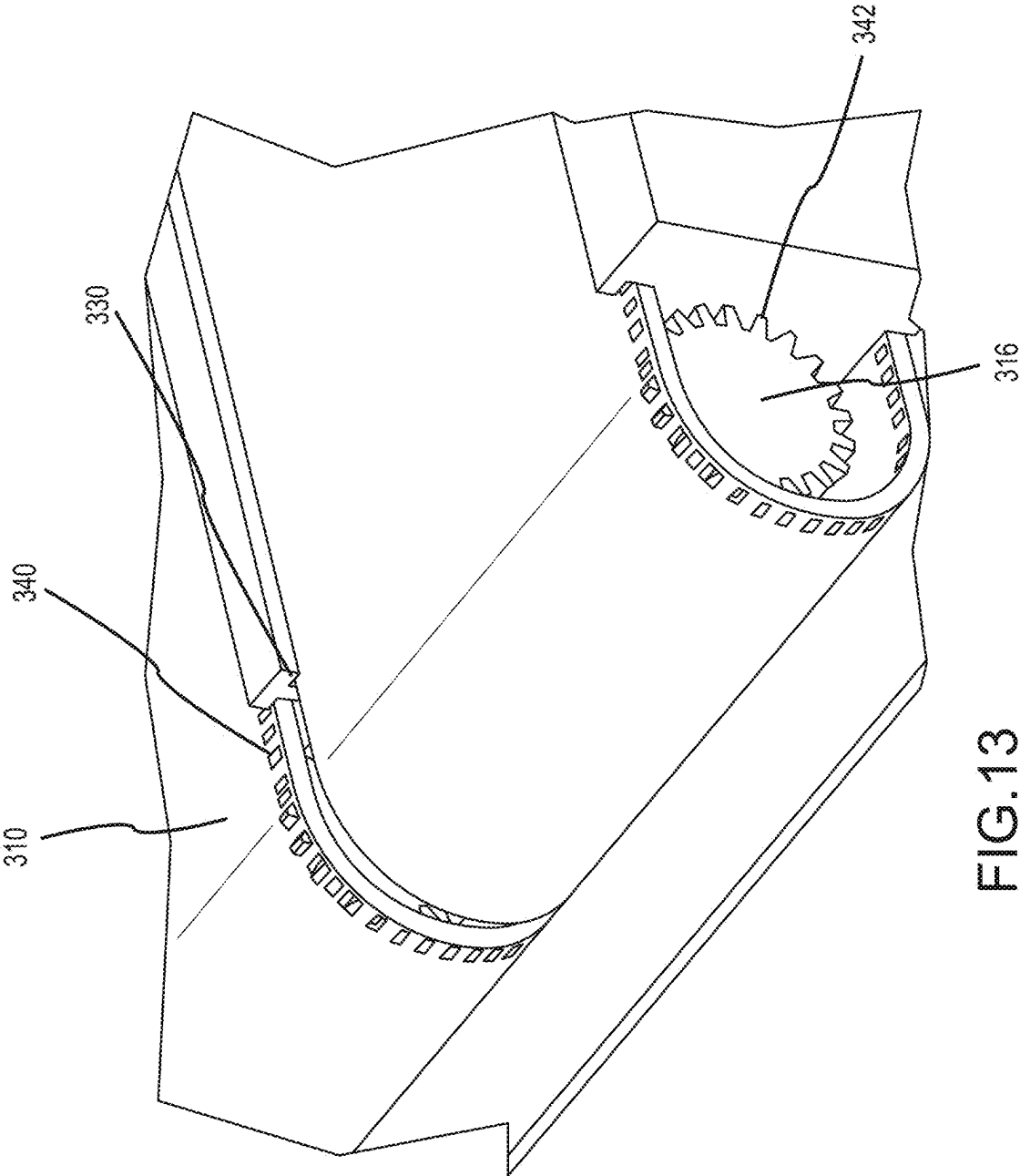


FIG.13

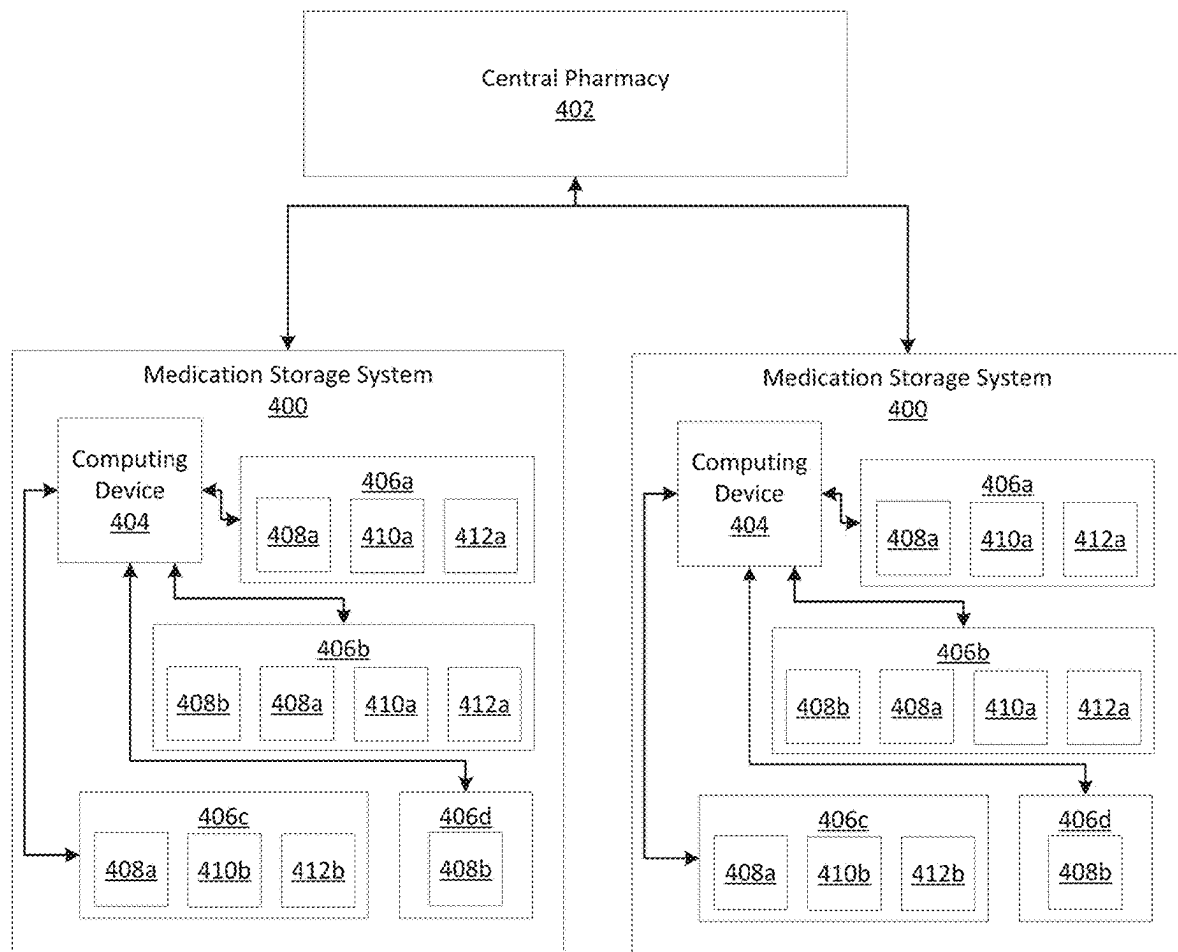


FIG. 14

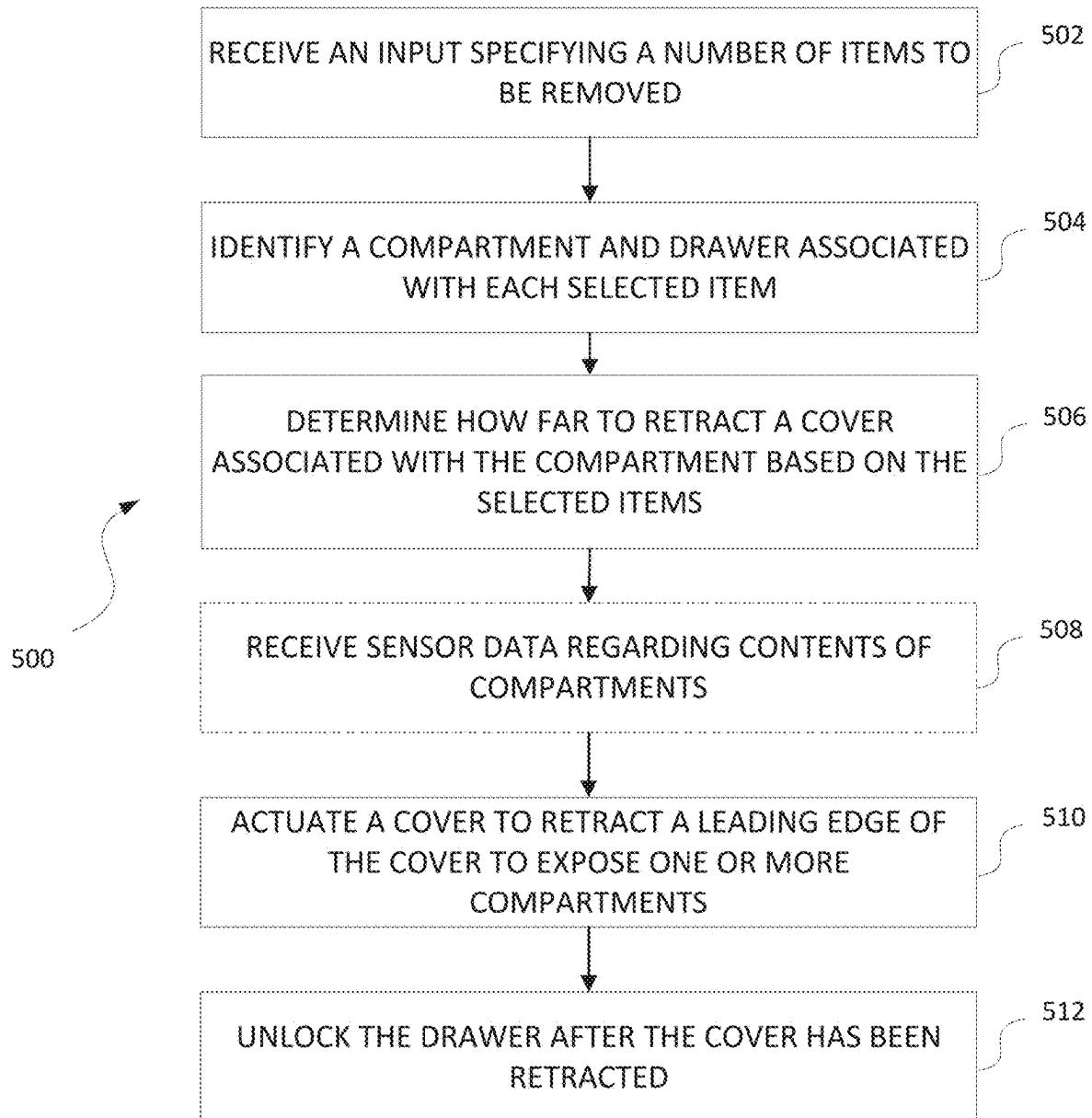


FIG. 15

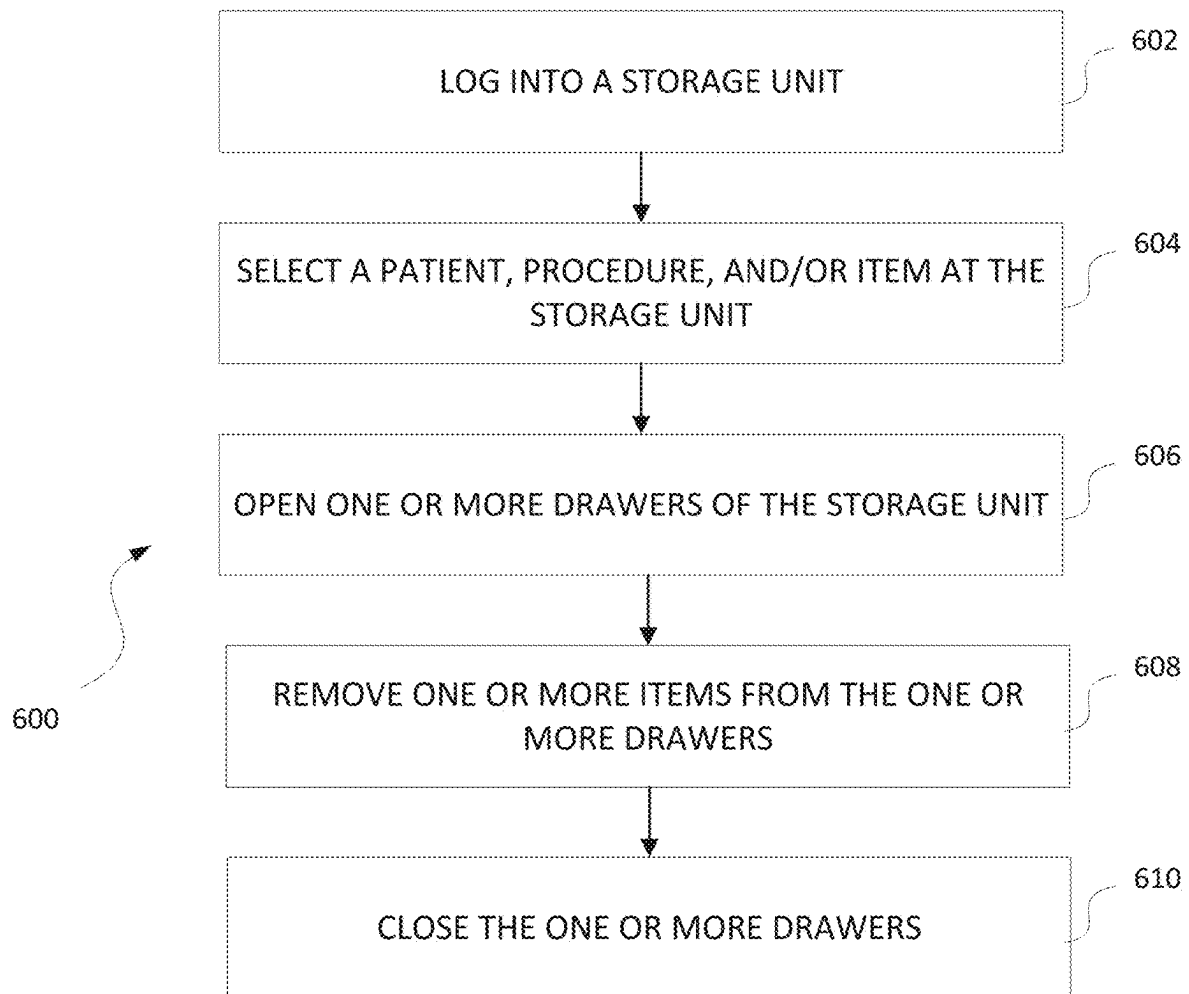
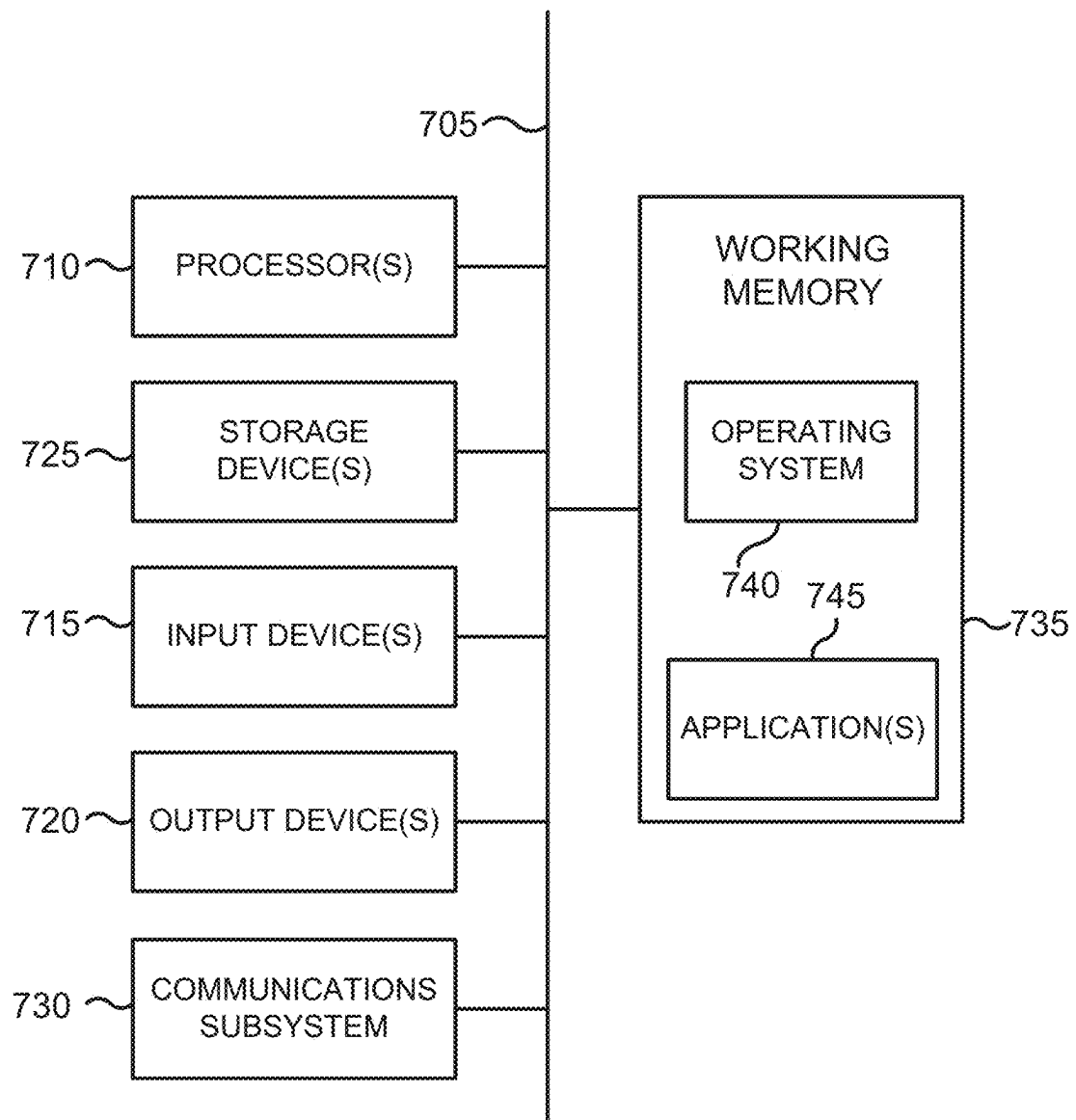


FIG. 16



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FIG. 17

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SENSOR DRIVEN SECURE DISPENSING UNIT**CROSS REFERENCE TO RELATED APPLICATIONS**

This application a continuation of U.S. patent application Ser. No. 18/166,051, filed Feb. 8, 2023, which is a continuation of U.S. patent application Ser. No. 16/834,950, filed Mar. 30, 2020, the complete disclosure of which is hereby incorporated by reference for all intents and purposes.

BACKGROUND OF THE INVENTION

Many industries rely on the accurate inventory and dispensing of secure items. For example, in a hospital setting, it is of paramount importance that patients be given the correct medications in the correct doses. In addition, it is legally required that controlled substances be securely stored and tracked to maintain proper levels of inventory and also to determine proper consumption. It is also important that inventories of medications and supplies be tracked so that proper business controls can be implemented.

Various dispensing cabinets and carts have been developed to assist in the management of medications and other items. However, improvements are still desired in the reliability of dispensing and tracking of items, and it is also desirable to reduce the amount of space required for item storage and dispensing.

BRIEF SUMMARY OF THE INVENTION

Embodiments of the present disclosure are directed to systems and methods for providing selectable controlled access to items stored within secured compartments. Embodiments may include a retractable cover that provides access to only the number of items selected by a particular user, without exposing any additional items or compartments. Additionally, embodiments of the present invention may help maintain inventory counts by keeping track of the actual presence of items, not just what the users say they are taking. This allows accurate inventory counts to be maintained in the event of user error and/or fraudulent behavior.

In one embodiment, a dispensing unit is provided. The dispensing unit may include a housing comprising a first end and a second end and a number of storage assemblies. The number of storage assemblies may be arranged in parallel with one another within the housing. Each storage assembly may include a number of compartments arranged linearly along an axis of the housing with each of the compartments being configured to store an item. The storage assembly may include a cover coupled with the housing and extending between the first end and the second end such that the cover is positioned over the compartments to secure items held therein. The storage assembly may further include an actuator coupled with the cover. The actuator may be configured to retract the cover to draw a distal end of the cover from the first end toward the second end to expose a selected number of the plurality of compartments.

In another embodiment, a dispensing unit may include a housing comprising a first end and a second end and a number of compartments arranged linearly along an axis of the housing, with each of the compartments being configured to store an item. The dispensing unit may also include a cover coupled with the housing and extending between the first end and the second end such that the cover is positioned over the plurality of compartments and an actuator coupled

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with the cover. The actuator may be configured to retract the cover to draw a leading edge of the cover from the first end toward the second end to expose a selected number of the plurality of compartments.

In another embodiment, a method of operating a dispensing mechanism is provided. The method may include receiving an input to dispense a selected number of items from a secure dispensing mechanism and actuating a cover of the secure dispensing mechanism to retract a leading edge of the cover to expose a selected number of compartments of the secure dispensing mechanism based on the selected number of items. The number of compartments may be arranged linearly along an axis of the secure dispensing mechanism, with each of the compartments being configured to store an item.

In another embodiment, a cabinet is provided. The cabinet may include a housing defining a storage region and at least one drawer. Each of the at least one drawer may include a first side and a second side and a plurality of compartments arranged linearly along an axis of the at least one drawer. Each of the compartments may be configured to store an item. The at least one drawer may also include a cover coupled with the housing and extending between the first side and the second side such that the cover is positioned over the plurality of compartments and an actuator coupled with the cover. The actuator may be configured to retract the cover to draw a leading edge of the cover from the first side toward the second side to expose a selected number of the plurality of compartments.

BRIEF DESCRIPTION OF THE DRAWINGS

A further understanding of the nature and advantages of various embodiments may be realized by reference to the following figures. In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If only the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label.

FIG. 1 illustrates an example cabinet in which the invention may be embodied.

FIG. 2 illustrates a drawer defining a number of compartments according to the present invention.

FIG. 3 illustrates a retractable cover interface with the drawer of FIG. 2.

FIG. 4 illustrates an embodiment of a drawer that may serve as a secure dispensing unit according to the present invention.

FIG. 5 illustrates a cover of the drawer of FIG. 4.

FIG. 6 illustrates a track to constrain movement of the cover of FIG. 5.

FIG. 7 illustrates an actuator of the cover of FIG. 4.

FIG. 8 illustrates a hub of the actuator of the cover of FIG. 7.

FIG. 9 illustrates a light sensor assembly of a compartment of the drawer of FIG. 4.

FIG. 10A illustrates an empty cassette according to embodiments of the present invention.

FIG. 10B illustrates the cassette of FIG. 10A in a filled state.

FIG. 11 illustrates a cassette that forms a portion of a row of compartments according to embodiments of the invention.

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FIG. 12 illustrates a mating assembly for mounting a cassette within a drawer according to embodiments of the invention.

FIG. 13 illustrates an alternative embodiment of a cover according to embodiments.

FIG. 14 is a block diagram of a medication control system according to embodiments.

FIG. 15 is a flowchart depicting a process for operating a secure dispensing mechanism according to embodiments.

FIG. 16 is a flowchart depicting a process for using a secure dispensing mechanism according to embodiments of the present invention.

FIG. 17 is a block diagram of a computing device in accordance with embodiments of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

The ensuing description provides exemplary embodiments only, and is not intended to limit the scope, applicability or configuration of the disclosure. Rather, the ensuing description of the exemplary embodiments will provide those skilled in the art with an enabling description for implementing one or more exemplary embodiments. It will be understood that various changes may be made in the function and arrangement of elements without departing from the spirit and scope of the invention as set forth in the appended claims. Merely by way of example, any embodiment described herein may or may not have any of the features discussed therewith, and may or may not have any feature discussed with respect to other embodiments.

Embodiments of the present invention are directed to dispensing mechanisms that provide controlled access to items stored therein. Embodiments of the invention include one more drawers or other storage units which each include a number of interior compartments that are usable to store one or more items, such as medications, tools, instruments, etc. Each drawer includes at least one cover that extends over at least some of the interior compartments. The cover may be retractable to provide selective access to a number of items and/or interior compartments. In some embodiments, the secure dispensing mechanism may be able to determine whether a particular compartment is empty and use this information to adjust a retraction distance of the cover to ensure access to the desired number of items is provided. For example, if a drawer includes a row of ten compartments that each include a single item and a user wishes to access three items, the dispensing mechanism may check to see if any of the compartments are empty by using one or more sensors, such as optical sensors, radio frequency (RF) sensors, load sensors, and the like. For example, if each of the ten compartments includes an item, the dispensing mechanism may retract the cover sufficiently far to provide access to the first three compartments, thereby allowing the user to remove the three items as requested. In another situation, the dispensing mechanism may determine that the first two compartments are empty and may then retract the cover sufficiently far to provide access to the first five compartments (including the two empty compartments), thereby allowing the user to remove the three items as requested.

Typically, the actuation of the cover is done while the drawer is in a closed position. By limiting the operation of the cover in such a manner, pinch points and/or other safety concerns are eliminated, as there are no moving parts once the drawer is open and available for user access. In operation, the user may interact with an interface of the dispensing

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mechanism to select a desired number of a particular type or types of items while the drawer is closed. The dispensing unit may then actuate one or more covers that correspond to a drawer and/or compartments within which the particular items are stored to retract each respective cover a proper distance to provide access to only those items that have been previously selected. Once the covers have been retracted, the relevant drawer or drawers may be unlocked, allowing the user to open the drawer and access the items within compartments which have been exposed by the retraction of the cover. After the items are removed, the user may shut the drawer. In some embodiments, the exposed compartments, now empty, may remain exposed, with the cover remaining partially retracted. In other embodiments, upon which the secure dispensing unit may lock the drawer and actuate the cover to extend the cover back over the previously exposed compartments. This may be particularly useful if some or all of the compartments include multiple items. For example, if only a single item is taken from a compartment having two items, the cover may be retracted to cover the remaining item. Oftentimes, during the retraction and/or extension of the cover, one or more optical sensors positioned on the cover may be used to monitor an inventory of items within the compartments.

Turning now to the drawings and referring first to FIG. 1, a cabinet 100 that may serve as and/or house a dispensing mechanism is illustrated (although it will be appreciated that the dispensing mechanisms may be embodied in non-cabinet forms, such as carts, built in shelving/drawers, and the like.). Here, cabinet 100 includes a storage area 102 that is used to store any number of items, such as tools, instruments, paints, medications, medical supply items, and the like. As illustrated, the storage area 102 defines an open interior that is configurable to receive a number of drawers 104 that may serve as dispensing units, including some drawers 104 that are secured storage units. In some embodiments, the use of other storage areas such as shelves, racks, and the like are possible in addition to, or as alternative options to, drawers 104. In some embodiments, the cabinet 100 may include one or more lockable doors (not shown) that control access to the storage area 102. Thus, a lockable door or doors could be provided in addition to, or as an alternative to, drawers 104. Drawers 104 may be of different sizes and shapes to perform different functions and/or to accommodate items of various sizes and/or shapes and/or to accommodate various equipment, such as sensors, security features, climate control equipment, and the like. As illustrated, different sizes of drawers 104 may be included in a single cabinet 100. In other embodiments, a cabinet 100 may include drawers 104 of uniform sizes. It will be further appreciated that a cabinet 100 may include drawers 104 of uniform function and/or interior configuration or may include one or more drawers 104 that have a different function and/or interior configuration than other drawers 104 in the cabinet 100.

In some embodiments, the cabinet 100 may include a work surface 106. The work surface 106 may provide an area for the user to set items on during preparation or completion of a particular procedure. For example, the user may lay out some or all of the items needed for the procedure on the work surface 106. In some embodiments, the work surface 106 may be a top surface of the cabinet 100, while in other embodiments, the work surface 106 may be positioned at an intermediate height of the cabinet 100. Additionally, while illustrated as forming substantially all of the top surface of the cabinet 100, it will be appreciated that

in some embodiments, the work surface **106** may not be present or may take up only a portion of the footprint of the cabinet **100**.

The cabinet **100** may also include and/or be in communication with a computing device **112**. The computing device **112** may include and/or be communicatively coupled with a display screen **114** and at least one input device. While illustrated with the display screen **114** coupled with the work surface **106** of the cabinet **100**, it will be appreciated that in some embodiments the display screen **114** may be integrated into a body of the cabinet **100** and/or positioned at other locations (such as extending laterally from or sitting atop a nearby structure) relative to the cabinet **100**. The input device **178** may include a keyboard, mouse, credential reader, microphone, camera and/or other device that enables a user to interact with the computing device **112**. In some embodiments, a credential reader may include a wireless reader, such as a Bluetooth, RFID, NFC, and/or other wireless reader that may read information from an active or passive user credential, such as a fob, mobile phone, ID, and/or other credential device. In other embodiments, the credential reader may include a contact reader, such as a chip or magnetic stripe reader. In yet other embodiments, the credential reader may include a biometric reader **180**, such as a camera or other optical sensor for facial, iris, and/or palm vein authentication, a microphone for voice authentication, a fingerprint reader, and/or other biometric sensor. In some embodiments, an input device may be incorporated into the display screen **114** by using a touchscreen display screen **114**.

The input devices of computing device **112** allows a user to interact with the cabinet **100**. For example, the user may interact with the input devices to log in, select a patient and/or procedure, and/or gain access to items stored within the cabinet **100**. For example, to log in to the cabinet **100**, a user may enter a user name, password, and/or other access credential (which may include a biometric credential) into the computing device **112** to verify that the user is authorized to access the cabinet **100** and/or a particular item stored therein. In other embodiments, a user may be logged in automatically if a wireless credential of the user is brought within range of a wireless credential reader of the computing device **112**. Once a user is logged into the computing device **112**, a graphical user interface (GUI) may be presented on the display screen **114** that allows the user to make selections about a patient, procedure, and/or items to be removed or otherwise dispensed from the cabinet **100**. These selections may be made using any of the input devices of the computing device **112**. In other embodiments, rather than navigating a GUI to make patient, procedure, and/or item selection, the user may use alternative selection means, such as voice commands, to make the necessary selections. In some embodiments, computing device **112** may include a network interface that enables the computing device **112** to communicate with a server and/or other remote computing device. The computing device **112** may use this connection to send an inventory of the cabinet **100** to the remote computing device. Such inventory counts may be updated periodically, such as every hour, work shift, day, etc. In other embodiments, the inventory may be tracked continuously before, during, and/or after each interaction with the cabinet **100**. In some instances, the central pharmacy may initiate a remote inventory scan of the cabinet **100** by sending a command to the computing device **112** that causes the computing device **112** to gather and/or send inventory information to the remote computing device. Such inventory information may be usable by the remote computing device to determine

when to refill and/or reorder a particular item. In some embodiments, the functions performed by computing device **112** could be performed by a remote server or other computing device over a network.

Once the user has selected which items are to be removed from the cabinet **100**, the cabinet **100** may provide access to the items, such as by unlocking a drawer **104** and/or otherwise providing access to the storage area **102**. Before, during, and/or after the user has removed any items from the cabinet **100**, the computing device **112** and/or other computer system may use any number of sensors to monitor which items were actually removed from the cabinet **100**. In some embodiments, the items that are actually removed are correlated with the items that the user originally selected for removal using the GUI. Such sensors may involve sensors integrated into the drawers **104** and/or those integrated into the work surface **106**. In some embodiments, if there is a discrepancy between selected items and the items removed, an indication of the discrepancy may be stored and/or sent to another device, as such behavior may be indicative of diversion behavior, improper use of the items, and/or other improper usage of the cabinet **100**. The indication may include information such as which user selected the items and which items were taken that did not match the selections.

Drawers **104** may be under the control of a computing device **112** and/or another controller. For example, each of drawers **104** may include an electronically controllable locking mechanism, and may only be openable under the control of computing device **112**. In some embodiments, some or all of the drawers **104** may be temperature and/or humidity controlled. In addition, computing device **112** may store information about what supplies are stored in which compartments of cabinet **100**. In one typical basic medical usage scenario, a health care worker may enter, using input device **178**, an identification of a patient who is under the care of the health care worker, and who will need medication during the worker's current rounds. Computing device **112** may access the patient's medical file and determine what medications have been prescribed for that patient. In other embodiments, a user may merely select one or more items to remove from the cabinet **100**, without the need to compare the data to an order, such as a patient treatment plan.

Once the items are selected, computing device **112** may unlock one or more drawers **104** (and/or doors, if used) that are associated with the selected items. In some embodiments, one or more lighted indicators may direct the health-care worker to a correct drawer **104**. A particular compartment within the correct drawer **104** may be highlighted, for example with a lighted indicator, to draw the user to the correct items. The user can then remove the selected items. In medical applications, the level of control exercised by computing device **112** may help in preventing medication and dosing errors, by reducing the likelihood that a health care worker will remove an incorrect medication from cabinet **100**. In addition, in all applications computing device **112** may document and record which items were dispensed, and may forward that information to inventory and/or accounting systems.

Each drawer **104** may define a storage area in which one or more items may be placed. In some embodiments, the storage area may be a large open space, while in other embodiments the storage area of drawer **104** may be divided into one or more compartments or bins. For example, as illustrated in FIG. 2, a single drawer **104** is shown. The drawer **104** may be adapted to fit the cabinet **100** as shown in FIG. 1. The drawer **104** may include a storage area **105**

that includes a number of bins or compartments **108** (which may or may not be formed as part of the drawer **104**). In some embodiments, the size of the compartments **108** may be uniform, while in other embodiments compartments **108** of different sizes and/or shapes may be included in a single drawer **104** to allow the drawer **104** to more efficiently store a number of different items. While in some embodiments each compartment **108** may be sized and/or shaped to hold only a particular item, it is possible in other embodiments that multiple items may be placed in a single compartment **108**. For example, multiple items may be stacked vertically, placed side by side, and/or placed top to bottom within a single compartment **108**. As illustrated, drawer **104** includes a number of compartments **108** arranged in rows, with all the compartments **108** in a given row being identical.

In some embodiments, drawer **104** may include a cover **110** that extends over at least some of the compartments **108** in a drawer **104**. For example, as illustrated in FIG. 3, a number of covers **110** are provided that extend across an entire row of compartments **108**. Here, the entire row of compartments **108** are oriented between front and back ends of the drawer **104**. In other embodiments, an entire row of compartments **108** may be oriented between the sides of the drawer **104**. In some embodiments, covers **110** extend across the entire row. In other embodiments one or more covers **110** may extend across any portion of a row of compartments **108**. In some embodiments, one or more covers **110** may extend over one or more of the compartments **108** in more than one row. In some embodiments, rather than having covers **110** atop every row of compartments **108**, one or more of the rows of compartments **108** may be open-topped such that a user may access contents of the open-topped compartments without the need to retract a cover **110**. Each cover **110** may be configured to be retracted to expose a selected number of compartments **108** and/or items. The cover **110** encloses the compartment **108** such that any item present within the compartment **108** may not be accessible to permit retrieval of the item. The cover **110** is moveable using suitable actuation mechanisms so that any particular compartment **108** may be uncovered to allow retrieval of an item stored within the compartment **108**. In one example, each of the one or more compartments **108** may be uncovered in a sequence, such that all compartments **108** that have been previously uncovered remain uncovered. In another example, the cover **110** may allow only one compartment **108** to be uncovered at any given time and for the rest of the compartments **108** associated with that cover **110** to be enclosed by the cover **110**. An actuator **116** (as shown in FIG. 2) may be coupled with each of the covers **110** such that the covers **110** may each be actuated to extend and/or retract the covers **110** independently of one another. In some embodiments, a single actuator may be used to move multiple covers **110**, while in other embodiments each cover **110** may include a dedicated actuator **116**. This ensures that if a user wants an item in a particular row of compartments **108**, only items from that row are exposed when the user accesses the drawer **104**. In addition to controlling access to non-selected items, by retracting covers **110** of only compartments **108** and/or items that are selected by the user prior to access, the user is able to quickly identify the items for removal, as the selected items will be the only ones that are not covered.

In embodiments that utilize drawers **104** with retractable covers **110**, once the user makes his selection of one or more items to retrieve from the cabinet **100**, the computing device **112** identifies one or more drawers **104** and/or compartments **108** that are associated with the selected items. The com-

puting device **112** may then send a signal to an actuator **116** associated with the relevant compartments **108**. The signal causes the relevant actuators **116** to activate to retract an attached cover **110** to expose a number of compartments **108** necessary to dispense the correct number of selected items. In some embodiments, multiple actuators **116** and covers **110** may be actuated in one session (either simultaneously or in sequence) to provide access to items in different rows and/or drawers **104**. Once the relevant covers **110** have been actuated, the relevant drawers **104** may be unlocked, allowing the user to access the drawers **104** and exposed compartments **108**. Once the user is done removing items, the user may close the drawer **104**. Once the computing device **112** detects that the drawers **104** are closed (such as by using one or more door close sensors) the computing device **112** may lock each drawer **104** and subsequently activate the actuators **116** to extend the cover **110** over the exposed compartments **108**. In such a manner, the cabinet **100** may prevent the possibility of any pinch points that may be created while the covers **110** are being moved, as the covers **110** are only moved when the drawers **104** are closed and/or locked.

In some embodiments, some or all of the drawers **104** may include one or more sensors that track the inventory and usage of items stored within the drawers **104** and/or compartments **108**. Such sensors may be specific to a particular drawer **104**, bin/compartment **108**, cassette, and/or portion thereof. In other embodiments, a single sensor and/or group of sensors may be used to monitor multiple drawers **104**, bins, and/or cassettes, and possibly an entire interior of the cabinet **100**. It will be appreciated that various arrangements and designs of drawers **104** may be used in conjunction with cabinet **100**.

Know the User

As discussed above, many applications of secure dispensing units, such as cabinet **100**, may require that a user provide authentication credentials in order to access the cabinet **100**. Several types of access credentials may be utilized. In some embodiments, logging in may be done by a user entering access credentials, such as a user name, password, and/or other credential. However, it may be desirable for a user to gain access quicker and/or easier than is achievable with manually keying in access credentials into a keyboard, keypad, and/or touchscreen. In such embodiments, other forms of access credentials may be utilized. For example, possession-based credentials may be utilized. In some embodiments, a possession-based credential may be in the form of a universal serial bus (USB) dongle, chip card, magnetic stripe card and/or other device that may be inserted into a port of a credential reader and/or otherwise scanned by a credential reader of the computing device **112**. Once inserted or read by the computing device **112**, the credential device may be authenticated to provide access to the patient, procedure, and/or item selection systems and may retrieve one or more items from the cabinet **100**. The authentication of the access credentials (possession-based, knowledge-based, and/or biometric) may be performed locally by the computing device **112** and/or may be performed remotely. When performed remotely, the authentication may involve the computing device **112** capturing the credentials and sending the credentials to a remote device, such as a backend server, for subsequent authentication. A result of the authentication of the credentials may be sent back to the computing device **112**, which will then control access to the functionality of the cabinet **100** based on the result.

In other embodiments, a possession based access credential may be in the form of a contactless device, such as a

radio frequency (RF) wireless device that may be wirelessly read by a credential reader of the computing device **112**. For example, the credential may be in the form of a card (such as an employee identification card), mobile phone, wristband, watch, other wearable, and/or other high integrity possession-based authentication object that may have an integrated RF chip. This enables the user to approach the cabinet **100** while in possession of one of these RF-enabled access credentials and gain access to the cabinet **100** with little to no log in action necessary by the user. For example, the RF-enabled access credential may include a Bluetooth® enabled device, RFID chip or tag, and/or other short range RF communication protocol that enables the access credential to be read by the credential reader of the computing device **112** as soon as the user is within a signal or detection range of the credential reader. In other embodiments, the access credential may operate using a shorter range communications protocol, such as near field communication (NFC). In such embodiments, the user may need to actively bring the access credential within signal range of an NFC credential reader of the computing device **112**.

In some embodiments, the RF chip of the access credential may be a passive chip that is powered by electromagnetic energy transmitted from an RFID reader/antenna of the credential reader of the computing device **112**. In other embodiments, the access credentials may include active RFID chips or tags that are powered by a battery (either a dedicated battery or a battery of a device containing the RFID chip or tag) continuously broadcast a signal containing the necessary access credentials for a particular user. Typically, such active RFID tags have a longer range at which the data can be read than the passive RFID tags. In some embodiments, the credential reader of the computing device **112** may be designed to have a predetermined signal/detection range that ensures that a user is sufficiently close to the cabinet **100** prior to reading a possession-based access credential. For example, a signal strength of the credential reader and/or RFID tag may be adjusted such that a desired signal range is achieved that helps prevent the computing device **112** from attempting to log in multiple users or an incorrect user when multiple people are positioned proximate the cabinet **100**. In other embodiments, the range of the cabinet **100** may be limited to a particular room or area in which the cabinet **100** is located. For example, RF shielding and/or other materials may be provided around a periphery of the room to ensure that only access credentials within the room may be detected by the credential reader of the cabinet **100**.

In some embodiments, rather than using a knowledge-based or possession-based access credential, the cabinet **100** may include one or more biometric readers that enable users to log in without carrying a physical access credential device. For example, the computing device **112** of cabinet **100** may include or be communicatively coupled with a fingerprint reader, a speaker for voice recognition, one or more optical sensors (such as a cameras, infrared (IR) scanners) for iris scanning, facial detection, palm vein recognition, and/or other biometric authentication techniques.

When using biometric authentication techniques that involve imaging a portion of a user (such as, but not limited to, facial recognition) anti-spoofing measures may be taken to help thwart fraudulent authentication attempts, such as when one user attempts to present a photograph, video, and/or mask of a different person who is an authorized user of the cabinet **100** to an image sensor of the cabinet **100**. Such anti-spoofing measures may include, for example,

active face liveness detection and/or passive face liveness detection. For active face liveness detection, the user may be asked to perform a specific action, such as nodding, blinking, smiling, and/or other facial pattern or gesture. If the computing device **112** determines that the user has performed the requested action, the computing device **112** may determine that the user being imaged is real and may be authenticated properly. In some embodiments, the facial pattern or gesture may be the same each time a user logs on, while in other embodiments, any number of facial patterns and/or gestures may be cycled through and/or randomly assigned to a particular login attempt. Using multiple facial patterns and/or gestures adds an additional layer of security that helps prevent videos from being displayed to the camera of the computing device **112** in an attempt to fraudulently login to the cabinet **100**.

Passive face liveness detection may involve various techniques. For example, some embodiments may utilize face flash liveness which uses a light element to illuminate the user (or copy of the user). The reflectance of the user is measured, allowing the camera to capture how the light from the screen and/or other light source reflects on the face. The computing device **112** can then determine whether the illuminated face belongs to a live person or a reproduction (such as a photo or a video) based on the measured reflectance. Some embodiments may perform passive face liveness detection using eye blink detection. For example, blinking by the user may be detected and timed. If no blinking is detected and/or the detected blinking is performed at a rate that is not common, the computing device **112** may deem the authentication attempt to be based on a reproduction of a user. Some embodiments may employ the use of trained convolutional neural networks (CNNs) to detect the authenticity of a user detected by a camera of the computing device **112**.

Additional passive anti-spoofing techniques may involve the use of infrared and/or 3-dimensional cameras, which may be able to readily distinguish between human users and reproductions such as 2-dimensional photographs and/or videos, as well as both 2-dimensional and 3-dimensional masks. For example, 3-dimensional cameras may be able to measure the depth of various features of an image and be able to distinguish between flat objects (e.g., photographs and/or display screens that may be showing a user's image) and 3-dimensional objects (such as a user or 3-dimensional mask). IR cameras (or other IR sensors) are capable of determining whether an image being detected has a thermal profile that matches that of a specific user and/or generally matches that of a human. For example, the thermal profile of a photo or display screen will not match that of a human. Similarly, both 2-dimensional and 3-dimensional masks will not have a thermal profile similar to a human unless a very sophisticated mask is utilized.

The use of voice recognition as an access credential typically involves a user speaking a particular word or phrase into a speaker of the cabinet **100**. The user's voice signature (tone, pitch, cadence, etc.) may be compared to a previously stored voice signature of known authorized users in order to detection where the user can be authenticated. In some embodiments, rather than having a user speak a same predetermined word or phrase, the cabinet **100** may prompt the user to speak a random one of a number of words or phrases in order to reduce the likelihood that someone could create an audio recording of an authorized user speaking a predetermined passphrase.

In some embodiments, multiple forms of credentials may be required to log in to further enhance the security of the

cabinet 100. For example, a physical access credential may be a chip card that requires the user to enter a personal identification number (PIN) in order to log in. Such embodiments require that the user have both possession of an access credential and knowledge of an alphanumeric login credential in order to be logged into the cabinet 100. In other embodiments, the cabinet 100 may require both a biometric credential (such as facial recognition) and a physical credential (such as a contactless credential device) in order to log into the cabinet 100. It will be appreciated that any combination of knowledge-based, possession-based, and/or biometric access credentials may be utilized to meet the security needs of a particular application. Additionally, in some embodiments, multiple types of a single type of credential may be utilized. For example, multiple forms of biometric credentials may be utilized (such as fingerprint and voice) to help further reduce the likelihood of fraudulent authentication.

Additionally, in some embodiments, backup authentication means (biometric and/or otherwise) may be assigned in the instance that a particular access credential is not available at a particular time. For example, if a user has a respiratory illness and cannot speak or can only speak with noticeable changes to his voice, an alternative to a voice recognition system (such as an alternative biometric credential, a knowledge-based credential, and/or a possession-based credential) may be utilized. Similarly, if a possession-based credential is not available, such as if an RF chip is not functioning properly or a user forgets a physical access credential, the cabinet 100 may provide an alternative authentication process that does not involve the unavailable access credential device. In some embodiments, a user may select an alternative authentication process, while in other embodiments such processes may be automatically provided by the cabinet 100 upon one or more failed attempts at successful authentication.

In one particular application, the cabinet 100 may be utilized as a cart for a physician, such as an anesthesiologist. Oftentimes, such personnel often have a need to access medications and/or equipment stored within a cabinet 100 quickly. Additionally, these users typically do not view the tracking of inventory as part of their duties. As a result, in such applications the cabinet 100 may be operated in a manner that ensures that the user may be logged on quickly and efficiently, with as little interaction as possible by the user. For example, in such embodiments the cabinet 100 may require hands free authentication credentials, such as possession-based wireless and/or contactless credential devices and/or biometric authentication credentials. Oftentimes, there may be a desired to utilize a contactless biometric authentication credential (i.e., not fingerprint) as such contact may be time consuming and/or less ergonomic than contactless biometric credentials. As a result, a cabinet 100 for a physician-based application may be configured to utilize facial, iris, palm vein, and/or voice credentials as login credentials. Additionally, hands free authentication credentials (possession-based and/or biometric-based) may be particularly useful in applications in which some or all of the users will be accessing the cabinet 100 with gloved hands, as gloves may make it more difficult for a user to key in access credentials, manipulate a contact-based credential device, and/or supply contact-based biometric credentials such as fingerprints. While discussed with using hands free authentication credentials for cabinet 100 that are accessed by physicians, it will be appreciated that in many applications authentication forms that require user actions and/or hand usage may be utilized in some embodiments.

Selecting an Item

Once a user is logged into the cabinet 100, the user may be prompted to enter additional details regarding what items are to be removed. For example, one or more items may be associated with a particular location and/or task. The selection may include what type and how many items are to be removed, possibly along with a particular task, location, and/or other information associated with the use of the item. This information, along with the knowledge of which user logged into the cabinet 110, allows the computing device 112 to track which user used each item and for what purpose. This may be particularly relevant in medical applications in which medications and/or other medical items are to be used in conjunction with a particular treatment and/or a particular patient. To enter these details, a user may interact with the computing device 112 using one or more input devices, such as a keypad, a keyboard, a mouse, a touchscreen display 114, and/or other input device. In some embodiments, the selection procedure may be voice controlled such that a user may select one or more items, tasks, patients, etc. Oftentimes, voice controlled systems may also include a manual entry selection system as a backup in case the voice control system is not functioning properly or in the event that a user has a respiratory illness or other cause of voice change or voice loss that may make it difficult to operate a voice controlled selection system.

In some embodiments for medical applications, a procedure list may be provided to the cabinet 100 that includes a limited number of patients, procedures (e.g., treatments), locations, personnel (nurses, techs, physicians, orderlies, etc.) to choose from. In some embodiments, the procedure list may be populated by scanning information from a patented wristband, chart, and/or other data source. In some embodiments, the procedure list may be manually populated at the cabinet 100 and/or at a remote computing device that then communicates the procedure list to the cabinet 100. In other embodiments, the procedure list may be automatically populated using a hospital (or other facility) scheduling system. For example, an electronic health records (EHR) system may be used to automatically populate the procedure list with data regarding specific patients and/or procedures that are currently on a schedule. In some embodiments, the EHR system may utilize knowledge about where a particular cabinet 100 is located in order to populate the procedure list. For example, if a cabinet 100 is located in an intensive care unit (ICU), the procedure list for the cabinet 100 may only be populated with only patients and/or procedures that correspond to patients present in the ICU, procedures performed in the ICU, and/or medical personnel staffing the ICU. Similarly procedure lists may be provided for each cabinet 100 in a designated area of a facility (NICU, maternity ward, operating room, etc.). In other embodiments, a cabinet 100 may include a procedure list that contains data about each patient and/or scheduled procedure in an entire facility. In other embodiments, the procedure list may be populated with any procedure that may be performed in the facility, regardless of likelihood or location of the cabinet 100. The procedure list for a given cabinet 100 may be presented on a GUI displayed on the display screen 114, allowing a logged in user to access the procedure list and make selections of items to access for one or more patients and/or procedures.

In some embodiments, a location of the cabinet 100 may be preprogrammed into the cabinet 100 (such as by tagging the cabinet 100 as being associated with a particular facility and/or portion thereof). In other embodiments, the EHR system and/or other central computing system may be

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programmed with a location of each respective cabinet **100** in a given facility. For example, a location may be associated with a serial number and/or other unique identifier of the cabinet **100** such that the EHR system has knowledge of the location of each cabinet **100** in a facility. In other embodiments, the location may be determined based on a wireless connection (such as a Bluetooth beacon) and/or other RFID tag that is usable to determine a location of each cabinet **100** in a facility and associate the location with an identifier of the respective cabinet **100**. In other embodiments, each cabinet **100** may include one or more other location determining features, such as a global positioning satellite (GPS) and/or other location sensor that may determine where in a facility the cabinet **100** is located.

In some embodiments, not only may a procedure list be populated based on a location and/or known patients and/or procedures, but the storage area **102** of the cabinet **100** may be populated with items, such as tools, medications, instruments, and the like based on the location and/or expected function of the cabinet **100**. For example, in a neonatal ward, only items and/or medications (and doses) that are relevant to the treatment of infants may be stored within the cabinet **100**. Similarly, a cabinet **100** in an operating room may be stocked with items and medications that may be used during surgical and/or other procedures. In some embodiments, such as where a procedure list is not used, is out of date, is incomplete, etc. a user may manually enter data about a patient and/or procedure into the computing device **112**.

Logged in users may interact with the procedure list (or other selection menu) to select item criteria (patient, case, expected procedure, location, medical personnel, etc.) using the GUI presented on the display screen. For example, the user may select and/or key in any necessary data and/or selections using a physical input device of the computing device **112** and/or may navigate the selection process using voice commands.

In some embodiments, rather than needing to select a patient and/or procedure, once a user is logged into the cabinet **100** only an item to be removed need be selected. This may be particularly useful in non-medical applications and/or other applications that do not involve patients and/or recipients of a particular item in which the task being performed with a particular item may not need to be taken into account. While patients and/or other recipients may not be involved, the cabinet **100** may oftentimes still require a selection of a task, project, and/or location associated with the use of an item. For example, if a tool is being removed from an cabinet **100** in a construction application, the user may need to select a particular project that the tool is being used on. This ensures that not only may the use of the item be attributed to the logged in user, but also that a specific task may be associated with the tool, which may better help the usage of the tool be tracked. This may also make tracking down lost items easier, as a last known location and/or project associated with the item may be known.

In some embodiments, once a user is logged into the cabinet **100**, the user may gain access to all of the contents within the storage area **102**. In other embodiments, one or more users of the cabinet **100** may have clearance levels that provide access to only a subset of the storage area **102**. For example, a nurse may have access to only non-controlled substances, while a physician may have access to an entirety of the storage area **102**, including controlled substances such as narcotics. In some embodiments, such controlled items may be housed in separate containers (such as high security drawers **104**) while in other embodiments controlled and non-controlled substances and/or other items may be stored

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within a single area, such as in a single drawer **104**. In some embodiments, to further enhance security of these controlled items when placed in drawers with less secure items, the controlled items may be placed in lockable bins that are secured within and/or otherwise provided within a drawer **104** and/or other feature of the storage area **102**. In some embodiments, the controlled items may be in compartments **108** that are secured by a retractable cover **110**. In other applications, each user may have access to items that pertain to their particular job function and/or training specialties. For example, if a user has not been certified to administer a particular form of treatment and/or perform a type of procedure, the user may not be given access to portions of the storage area **102** that contain items that are used only in the particular treatment and/or procedure.

Access

Once a user has made selections about any items to be retrieved from the cabinet **100** (including any other data, such as patients, procedures, tasks, locations, personnel, etc.), the cabinet **100** may provide access to the interior of the storage **102** to allow the user to take the items that the user is authorized to possess and/or administer. In some embodiments, this may be achieved by the computing device **112** sending an unlock command to one or more drawers **104** positioned within the storage area **102**. The unlock command may cause a locking mechanism, such as a solenoid-actuated lock, to disengage and allow the drawer **104** to be opened. In some embodiments, a cover **110** within a drawer **104** may be retracted to expose one or more compartments **108** to provide access to controlled items (such as narcotics).

Once the user gains access to the storage area **102** and may take the selected items. In some embodiments, multiple drawers **104** and/or other storage units may need to be accessed by a user to retrieve all of the items that have been selected for use. Once the items have been taken, the user may close the drawer **104** and/or other portion of storage area **102**. The drawers **104** and/or other partitions may be locked again by the cabinet **100** to secure any remaining items within the storage area **102**. Any retracted covers **110** may then be extended to again cover the exposed compartments **108**.

Turning to FIG. 4, an embodiment of a drawer **204** that may serve as a secure dispensing unit is illustrated. In some embodiments, drawer **204** may be similar to drawer **104** and/or may be used in cabinet **100** described above. For example, drawer **204** include a storage area **206** that include a number of compartments **208a** (which may or may not be formed as part of the drawer **204**). While illustrated with the size of the compartments **208a** being uniform, different sizes and/or shapes may be included in a single drawer **204** to allow the drawer **204** to more efficiently store a number of different items. Oftentimes, each compartment **208a** may be sized and shaped to hold only a single item, however in some embodiments multiple items may be stacked, placed side by side, and/or placed top to bottom in a single compartment **208a**. As illustrated, drawer **204** includes a number of compartments **208a** arranged in rows, with all the compartments **208a** in a given row being identical.

In some embodiments, drawer **204** may include a cover **210** that extends over at least some of the compartments **208a** in a drawer **204**. Here, each row of compartments **208a** includes its own dedicated cover **210** that is configured to extend across an entire row of compartments **208a**. Each cover **210** may be configured to be retracted to expose a selected number of compartments **208a** and/or items and to be extended to recover the compartments **208a** when access

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is no longer needed. The cover **210** may be a single piece of flexible material and/or multiple pieces of flexible and/or rigid material that are coupled together to form a length of material having a first end and second end. When decoupled from the drawer **204**, the cover **210** may be extended into a single planar strip of material with the first end and second end at opposite ends of the length of the strip of material. The cover **210** is sufficiently strong to prevent users from accessing covered compartments **208a**. In some embodiments, the cover **210** may be opaque to prevent covered compartments **208a** from being viewable, while in other embodiments the cover **210** may be transparent to allow users to view the contents of covered compartments **208a**.

As indicated above, the cover **210** may be formed from a number of pieces of material. As best illustrated in FIG. 5, a number of segments **222** may be linked together similar to a watch band. For example, each segment **222** may include a female portion **224** at one end and a male portion **226** at the other end. As just one example, the female portion **224** may include a central recess formed between two extensions. The male portion **226** may include a central extension surrounded by two indented portions. The central recess of the female portion **224** may be slightly larger than the central extensions of the male portion **226** such that the central extension of the male portion **226** of a first segment **222** may be inserted into the central recess of the female portion **224** of a second segment **222**. The extensions of the two segments **222** may then be pivotally coupled together to link the segments **222**. For example, a pin **262** may be inserted through each segment to link the extensions of the segments **222** together. In such a manner, a number of segments **222** may be linked together in a manner that allows the linked segments **222** lengths of material that may be arranged in planar formations to cover the compartments **208a** and/or curved to follow a curved movement path as the cover **210** is extended and/or retracted. In some embodiments, a portion of the cover **210** may be arranged in a planar manner while another portion of the cover **210** has segments **222** that are pivoted relative to one another to curve a portion of the cover **210**. It will be appreciated that other designs of pivotable segments may be used to create the cover **210**.

In some embodiments, a cover may be constrained to a particular movement path via one or more tracks formed in a given compartment. For example, as illustrated in FIG. 6, tracks **218** are formed in a compartment **290**, which may be similar to compartment **208** described herein. The tracks **218** may be provided near top edges of each compartment **290** and/or row of compartments **290** that a cover (such as cover **210**) is intended to secure. As illustrated, the tracks **218** are in the form of channels that receive lateral edges of the cover to ensure that the cover moves linearly along a longitudinal axis of the cover to cover and expose the compartments **290** and/or rows. The tracks **218** are positioned proximate a top of the compartments **290** so as to effectively seal compartments **290** that are below a position of the cover. While not shown, in some embodiments tracks **218** may be provided below the compartments **290**, providing a movement path below the compartments **290** that enables the cover **210** to be drawn back and rolled under the compartments **290** in a manner similar to a roll top desk. In embodiments in which the cover is formed of linked segments (such as linked segments **222**), as the cover is retracted and rolled back, some of the segments may be pivoted to curve and wrap around to move underneath the compartments **290**.

An actuator **216** may be coupled with each of the covers **210** such that the covers **210** may each be actuated to extend and/or retract the covers **210** independently of one another

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as best illustrated in FIG. 7. Oftentimes, the actuator **216** may include a motor **220** that is configured to generate force that may be applied to the cover **210**. In some embodiments, the actuator **216** may include a transmission device that transfers the force from the motor **220** to the cover **210**. For example, as illustrated in FIG. 8, the actuator **216** may include a hub **228** that coupled with edges of the cover **210**. As illustrated, hub **228** includes a number of cogs **229** that are distributed radially about the outer periphery of the hub **228**. In some embodiments, the pins **262** used to connect each segment **222** may extend laterally beyond the edges of the segments **222**. Ends of each pin **262** may fit in spaces **231** between the cogs **229** of the hub **228**, allowing the rotation of the motor **220** and hub **228** to retract and extend the cover **210** based on the direction of the rotation. In other embodiments, edges of the cover **210** (such as edges of each segment **222**) may define apertures (not shown) that are configured to receive the cogs **229** of the hub **228** as the hub **228** is rotated by the motor **220**.

In some embodiments, the motor **220** may include and/or be coupled with an encoder (not shown) that informs the motor **220** how many compartments **208** have been exposed. For example, the encoder may include information associated with a size of each compartment **208**, a rotational speed of the motor **220**, an effective radius of the transmission, and/or other information that may be usable to determine how far the cover **210** has been moved relative to the compartment **208**. This ensures that the encoder is capable of allowing the motor **220** to actuate to move the cover **210** to positions in which a distal end of the cover **210** lines up substantially with boundaries of the respective compartments **208**. This ensures that exactly the correct number of compartments **208** and/or items are exposed when the user accesses the drawer **204**. Additionally, by ensuring that the covers **210** are retracted to precise positions, the user is able to quickly identify the correct items and correct number of items for removal without any questions of whether to remove items that are in partially exposed compartments **208**, as the selected items will be the only ones that are exposed.

In operation, once the user makes a selection of one or more items to retrieve from the cabinet **100**, the computing device **112** identifies one or more drawers **204** and/or compartments **208** that are associated with the selected items. The computing device **112** may then send a signal to an actuator **216** associated with the relevant compartments **208**. The signal causes the relevant actuators **216** to activate to retract an attached cover **210** to expose a number of compartments **208** necessary to dispense the correct number of selected items. In some embodiments, multiple actuators **216** and covers **210** may be actuated in one session (either simultaneously or in sequence) to provide access to items in different rows and/or drawers **204**. Once the relevant covers **210** have been actuated, the relevant drawers **204** may be unlocked, allowing the user to access the drawers **204** and exposed compartments **208**. Once the user is done removing items, the user may close the drawer **204**. Once the computing device **112** detects that the drawers **204** are closed (such as by using one or more door close sensors) the computing device **112** may lock each drawer **204** and subsequently activate the actuators **216** to extend the cover **210** over the exposed compartments **208**. In such a manner, the cabinet **200** may prevent the possibility of any pinch points that may be created while the covers **210** are being moved, as the covers **210** are only moved when the drawers **204** are closed and/or locked.

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In some embodiments, the actuators **216** may include one or more manual override mechanisms. Such mechanisms may allow the cover **210** to be moved manually in certain events, such as due to the occurrence of a power failure. For example, the actuator **216** may include a clutch or coupler that is disengageable from the actuator **216** to enable the user to move the cover **210** independently of the actuator **216**. For example, a mechanical and/or electrical key lock may be used to enable the user to manually move the cover **210** to access the items within the compartments **208**. In some embodiments, the actuator **216** may include a slot that accepts a winding key that the user may utilize to wind and/or otherwise manually manipulate the actuator **216** if access is needed and there is no power. In some embodiments, the actuator **216** may include a gearbox and/or a locking pawl that prevents the user from manually moving the cover **210** in normal operation.

Oftentimes, one or more of the compartments **208** may be empty prior to its respective cover **210** being retracted. For example, one or more users may have previously accessed a particular row of compartments **208** since the last time drawer **204** was refilled. In such embodiments, the drawer **204** may include a number of sensors that help detect whether the compartments **204**, drawer **204** are empty or contain an item. Based on this information, the cover **210** may be actuated to expose a number of compartments **208** that allows access to the selected number of items. For example, if a drawer **204** includes a row of ten compartments **208** that each include a single item and a user wishes to access three items, the computing device **112** may check to see if any of the compartments **208** are empty by using one or more sensors. For example, if each of the ten compartments **208** includes an item, the computing device **112** may retract the cover **210** sufficiently far to provide access to the first three compartments **208**, thereby allowing the user to remove the three items as requested. In another situation, the computing device **112** may determine that the first two compartments **208** are empty and may then retract the cover **210** sufficiently far to provide access to the first five compartments **208** (including the two empty compartments **208**, thereby allowing the user to remove the three items as requested.

In some embodiments, one or of the sensors used determine whether a particular compartment **208** is full or empty may be integrated into the cover **210** itself. For example, a leading edge **230** of the cover **210** may include one or more optical sensors **232** that may be used to detect any contents of the compartments **208** as the cover **210** is retracted and/or extended. In some embodiments, the leading edge **230** may be formed from a last segment **222** of the cover **210**. By placing the optical sensor **232** in the leading edge **230**, the computing device **112** is able to determine whether each compartment **208** is empty as the cover **210** is retracted, starting with a distal most compartment **208** over which the cover **210** extends. In some embodiments, the optical sensors **232** (such as cameras, infrared sensors, and the like) may be configured to image the contents of a compartment **208** to determine whether one or more items are stored therein. For example, the computing device **112** may perform object detection on images captured by the optical sensor **232** of each compartment **208** as the cover **210** is extended and/or retracted. The object detection may be usable to determine whether each compartment **208** is empty or contains one or more items.

In other embodiments, the optical sensors **232** may work in conjunction with one or more features of the compartments **208** to determine whether any items are present in the

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compartment **208**. For example, in some embodiments, a computer-readable identifier (such as a barcode, quick response (QR) code, alphanumeric text, and/or other object that is readable by a computer) may be printed, adhered, and/or otherwise placed on a bottom surface of each compartment **208**. If an item is positioned within the compartment **208**, the item obscures all or part of the computer-readable identifier, rendering the identifier unreadable. Thus, if the optical sensor **232** cannot read the identifier of a particular compartment **208**, the computing device **112** may determine that the compartment **208** is housing an item. If the optical sensor **232** is able to read the identifier, the computing device **112** may conclude that the compartment **208** is empty.

In another embodiment, a light source, such as a light emitting diode (LED) may be provided within a base of each compartment **208**. Optical sensor **232** within the leading edge **230** may be configured to detect the presence of any light emitted from the light source. In some embodiments, the optical sensor **232** may be a receiver that is configured to react to light signals that are pulsed at a particular frequency. In some embodiments, the optical sensor **232** may be in the form of a phototransitive receiver that is activated when it detects light emitted at a particular wavelength. In some embodiments, the light may include wavelengths in the IR spectrum, although other wavelengths are possible in some embodiments. If the light is detected by the optical sensor **232**, the computing device **112** may determine that the compartment **208** is empty as no object is interfering with the ability of the optical sensor **232** to detect the light. Similarly, if the optical sensor **232** is unable to detect the light, the computing device may determine that the compartment **208** has an item stored therein that is blocking the light. In some embodiments, the optical sensor **232** may detect only a portion of the light emitted by the light source. In such embodiments, the computing device **112** may be programed based on the needs of a particular application. For example, in some embodiments, if the optical sensor **232** detects only a portion of the light emitted from a light source, the computing device **112** may determine that an item is blocking the rest of the light and that the compartment **208** must be occupied. In other embodiments if any light is detected the computing device **112** may determine that the compartment **208** is empty.

In some embodiments, the light source may be positioned directly in a storage area of the compartment **208**, such as by placing a LED or other light source in a base **209** of the compartment **208** directly below where an item is to be stored. In other embodiments, such as that illustrated in FIG. 9, the light source **211** (such as a LED) may be positioned outside of the compartment **208**. In some embodiments, such as if the compartment **208** is removable from the drawer **204**) the light source **211** may be affixed to a device other than the compartment **208**. For example, the light source **211** may be positioned on a housing of the drawer **204**. In order to introduce the light into the interior of the compartment **208**, an aperture **213** may be formed in a bottom and/or side of the compartment **208** that is in the illumination field of the light source **211**. In some embodiments, a light pipe **234** and/or other structure may be provided in the base **209** of the compartment **209** to further distribute the emitted light through the compartment **208**. As illustrated, the light pipe **234** may be coupled with a portion of the base **209** of the compartment **208** and extends through a sidewall **215** of the compartment **208**, serving to direct light from an externally located light source **211** to be introduced into the interior of the compartment **208**. Oftentimes, the light pipe **234** is

positioned in a central region of the compartment **208**, which may help ensure that if an item is present within the compartment **208** then the light and light pipe **234** will be obstructed from detection by the optical sensor **232**.

In some embodiments, the compartment **208** and/or light pipe **234** may be designed to further ensure that items present within the compartment **208** will obstruct the light from being detected by the optical sensor **232**. For example, as illustrated, the light pipe **234** may only extend along a portion of the length of the compartment **208**, such that no portion of the light pipe **234** extends beyond a distal end **217** of an item placed within the compartment **208**. Additionally, the base **209** of each compartment **208** may be sloped along one or more axes **219a**, **219b** toward the light pipe **234**, thereby biasing items stored within the compartment **208** to be positioned directly above the light pipe **234** in order to obstruct the light emitted from the light pipe **234**. For example, lateral sides **221** of the base of each compartment **208** may be curved and/or otherwise tapered toward light pipe **234** positioned in the center of the compartment **208**. Additionally, the base **209** of each compartment **208** may be sloped from one longitudinal end opposite the light source toward the other longitudinal end through which the light source **211** and light pipe **234** extend. Not only does the tilting of the base **209** of the compartments **208** help ensure that items placed within the compartment **208** are positioned to cover the light pipe **234** to increase the accuracy of optical sensors **232**, but also helps to make items easier to remove from the compartments **208**.

In some embodiments, non-optical sensors may be used instead of or in conjunction with optical sensors to determine whether a particular compartment **208** is empty. For example, each compartment **208** may include one or more load sensors positioned in or below a base of the compartment **208**. These load sensors (such as strain gauges, capacitive sensors, and the like) may be calibrated to provide non-zero readings when an item is present. In some embodiments, a measurement of a load sensor may be compared to a known weight of a particular item that is expected to be stored in a given compartment **208**. If the load measurement matches the known weight within an error factor of the load sensor, the computing device **112** may determine that the compartment **208** associated with the load sensor is housing an item.

In another embodiment, each item within a compartment **208** may include an RF tag (passive or active) that may be read by an antenna disposed on the leading edge **230** of the cover **210** as the cover **210** is extended and/or retracted. In such embodiments, RF shielding may be provided on the walls and base of each compartment **208** such that only items within a single compartment **208** may be read at a single time by the antenna. It will be appreciated that other sensors may be integrated into the cover **210**, compartment **208**, and/or drawer **204** to track the presence and usage of items positioned within the compartments **208**.

In addition to helping determine the filled status of a compartment **208** to actuate the cover **210** a correct distance, the information from sensors may be used to help monitor an inventory of the drawer **204**. For example, after the cover **210** is retracted and/or extended, the computing device **112** may monitor how many compartments **208** have items stored therein based on data from the various sensors. This information may be tracked, oftentimes along with information about which users took the items and possibly for what purpose (task, patient, etc.) to monitor how many items are present within the drawer **204** and/or cabinet **100** and how items have been utilized. In some embodiments, the

inventory and/or usage data may be updated each time the covers **210** are extended and/or retracted and/or when each drawer **204** is opened and/or closed.

In some embodiments, compartments may be formed as part of the drawers **204** themselves. In other embodiments, such as illustrated here, the compartments **208** may be separately formed and subsequently coupled with a base of the drawers **204**. For example, each compartment **208** may be formed as an individual bin that is later coupled with a base of the drawer **204**. In other embodiments, such as that shown in FIGS. **10A** and **10B**, each compartment may be formed as part of a cassette **236** and/or other removable feature that includes a number of compartments **208**. While illustrated with ten uniformly sized compartments **208**, it will be appreciated that cassette **236** may include any number and size of compartments **208**. Additionally, while arranged in a single row, it will be appreciated that in some embodiments a cassette **236** may include multiple rows of compartments **208**. In some embodiments, each cassette **236** may form an entire row of compartments **208** within a drawer **204**. In other embodiments, a cassette **236** may form only a portion of a row of compartments **208**, such as one half or other fraction of a row of compartments **208**. Such a configuration provides benefits associated with refilling the drawer **204**. For example, if the compartments **208** in a given row are divided into two cassettes **236** as shown in FIG. **11**, a refill alert may be generated when the first half of the row is empty, ensuring that the refill occurs while the second half of the row is still has items available to a user. To refill the first half, the user may use the computing device **112** to select a refill procedure. This may cause the computing device **112** to retract the cover **210** to expose at least the first half of the row so that the empty one of the cassettes **236** may be removed and replaced with a full cassette **236**. In other embodiments, a drawer **204** may be refilled on a compartment by compartment basis. In such embodiments, the cover **210** may be retracted to expose all empty compartments **208** and a user and/or pick and place robot may insert the correct items into the empty compartments **208**. In some embodiments, the cassettes **236** may include covers (not shown) for transporting full cassettes **236**. For example, once a cassette **236** has been filled by a pharmacy, a cover (such as a reusable cover, a cover similar to cover **110**, a one-time use cover (such as a tape or other film), and/or other cover) may be positioned atop the various compartments **208** within the cassette **236**. Before, during, and/or after the cassette **236** is inserted into a drawer **204**, the cover may be removed such that the cover **210** of the drawer **204** may be used to control access to the various items within the compartments **208**.

As illustrated, each compartment **208** within the cassette **236** may include light pipe **234**. The light sources may be coupled with the compartments **208** and/or disposed within the drawer **204** such that when the cassette **236** is mounted within the drawer **204** the light sources are in alignment with the light pipes **234** of the individual compartments **208**. In some embodiments, other sensors, such as load sensors, RF sensors, and the like may be coupled with the cassette **236** and/or with the drawer **204** itself. In some embodiments, each cassette **236** may include an identifier (such as an identifier that is encoded onto an RF chip) that indicate which items and/or drawer **204** are associated with the given cassette **236**.

In some embodiments, each cassette **236** may include the cover **210** and actuator **216** assembly, while in other embodiments, the cassettes **236** may be inserted into the cover **210** and actuator **216** assembly. For example, the cover **210** and

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actuator **216** assembly may be coupled with the drawer **204** such that to insert a cassette **236** the cover **210** must be at least partially retracted. In some embodiments, an underside of each of the cassettes **236** may include one or more mating features, such as clips **238**, buttons, magnets, and/or other coupling mechanisms that are configured to secure the cassette **236** to the base **241** of the drawer **204**. As illustrated in FIG. **12**, the base **241** of the drawer **204** may define one or more mating features **240** (such as recesses and/or other structures that may serve as mounting locations) that are configured to engage with the clips **238** and retain the cassette **236** within the drawer **204**. In some embodiments, the connection of mating features (such as the engagement of clips **238** into the mating features **240**) of the cassettes **236** and the drawers **204** may be similar to those in ski boots, where a front end of a connector is inserted into a mating feature and the rear end is forced downward to lock the cassette **236** in place.

FIG. **13** illustrates an alternative embodiment of a cover **310**. Cover **310** may operate in a manner similar to cover **210** described above. Here, cover **310** is a formed of a single piece of flexible material (such as sheet metal, polymers, and the like) that is capable of being curved about an actuator and/or drive component coupled with the actuator (such as gear **316**) as the cover is retracted and/or extended. Lateral edges of cover **310** define a number of apertures **340** that are configured to receive cogs **342** of the actuator **316**, allowing the actuator **316** to propel the cover **310**. In some embodiments, the motion of cover **310** is constrained by a track **330** formed above and/or below individual compartments of a drawer.

FIG. **14** illustrates an embodiment of a medication control system **414**. In one particular embodiment, a medication control system **414** may be provided in a medical facility, such as a hospital or clinic. In such embodiments, the medication control system **414** may include one or more medication storage systems **400** that are in communication with a central computing device, such as a central pharmacy **402**. Each medication storage system **400** may be embodied in any form, such as a cabinet (similar to cabinet **100**), cart, shelf, locker, and/or other storage unit and may be positioned in a particular room, floor, and/or ward of the facility.

Each medication storage system **400** may include a computing device **404** (similar to computing device **112**) that is configured to communicate information between the medication storage system **400** and the central pharmacy, authenticate users of the medication storage system **400**, and control access to medications and/or other items stored within the medication storage system **400**. The computing device **404** may include and/or be communicatively coupled with a display screen and at least one input device, such as a keyboard, mouse, touchscreen, credential reader, microphone, camera and/or other device that enables a user to interact with the computing device **404**. The credential reader may be in the form of a wireless reader, such as a Bluetooth, RFID, NFC, and/or other wireless reader that may read information from an active or passive user credential, such as a fob, mobile phone, ID, and/or other credential device. In other embodiments, the credential reader may include a contact reader, such as a chip or magnetic stripe reader. In yet other embodiments, the credential reader may include a biometric reader, such as a camera or other optical sensor for facial, iris, and/or palm vein authentication, a microphone for voice authentication, a fingerprint reader, and/or other biometric sensor.

A user is able to interact with the computing device **404** using one or more of the input devices to log in, select a

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patient and/or procedure, and/or gain access to items stored within the medication storage system **400**. For example, to log in to the medication storage system **400**, the user may enter a user name, password, and/or other access credential (which may include a biometric credential) to verify that the user is authorized to access the cabinet **100** and/or a particular item stored therein. In other embodiments, a user may be logged in automatically if a wireless credential of the user is brought within range of a wireless credential reader of the computing device **404**. Once a user is logged in, the user may make selections about a patient, procedure, and/or items to be removed or otherwise dispensed from the medication storage system **400**. These selections may be made using any of the input devices of the computing device **112**. In one typical basic medical usage scenario, a health care worker may enter an identification of a patient who is under the care of the health care worker, and who will need medication during the worker's current rounds. Computing device **404** may access the patient's medical file and determine what medications have been prescribed for that patient. In other embodiments, a user may merely select one or more items to remove from the medication storage system **400**, without the need to compare the data to an order, such as a patient treatment plan.

Once the user has selected which items are to be removed, the medication storage system **400** may provide access to the items, such as by the computing device **404** unlocking one or more drawers **406** (similar to drawers **104** and **204** as described above) that are configured to store various medications and/or other items. Before, during, and/or after the user has removed any items from the cabinet **100**, the computing device **404** and/or other computer system may use any number of sensors to monitor which items were actually removed from the medication storage system **400**. The drawers **406** may be independently lockable such that access to each drawer **406** may be controlled individually. In some embodiments, a medication storage system **400** may include drawers **406** of a single configuration. In other embodiments, such as illustrated here, the medication storage system **400** may include drawers **406** of different layouts and/or sizes.

For example, as illustrated, the medication storage system **400** includes four drawers **406** that each have a different size and/or drawer layout, although it will be appreciated that any number of combinations of drawer sizes and/or layouts is possible in some embodiments. Here, drawer **406a** defines a storage region that has been separated into a number of bins and/or other compartments **408a** for receiving one or more items. The compartments **408a** may be arranged in rows and/or columns within the interior of the drawer **406a**. Compartments **408a** are configured to be covered by a retractable and/or otherwise removable cover **410a**. Cover **410a** may be similar to covers **210** and **310** described above. For example, a number of covers **410a** are provided that extend across an entire row of compartments **408a**, which may be oriented across a width of the drawer **406a**. Oftentimes, covers **410a** extend across the entire row of compartments **408a**. In other embodiments, one or more covers **410a** may extend across any portion of a row of compartments **408a**. Each cover **410a** may be configured to be retracted to expose a selected number of compartments **408a** and/or items. The cover **410a** encloses the compartment **408a** such that any item present within the compartment **408a** may not be accessible to permit retrieval of the item. The cover **410a** is moveable using an actuator **412a** (similar to actuator **116** and **216** described above) that moves the cover **410a** to uncover an item stored within the compartment **408a**. For

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example, the actuator **412a** may be coupled with each of the covers **410a** such that the covers **410a** may each be actuated to extend and/or retract the covers **410a** independently of one another.

In one example, each of the one or more compartments **408a** may be uncovered in a sequence, such that all compartments **108** that have been previously uncovered remain uncovered. In another example, the cover **410a** may allow only one compartment **408a** to be uncovered at any given time and for the rest of the compartments **408a** associated with that cover **410a** to be enclosed by the cover **410a**. This ensures that if a user wants an item in a particular row of compartments **408a**, only items from that row are exposed when the user accesses the drawer **406a**. In addition to controlling access to non-selected items, by retracting covers **410a** of only compartments **408a** and/or items that are selected by the user prior to access, the user is able to quickly identify the items for removal, as the selected items will be the only ones that are not covered.

As illustrated, medication storage system **400** includes another drawer **406b**. Here, drawer **406b** includes both covered compartments **408a** and open-topped compartments **408b**. Covered compartments **408a** include covers **410a** and actuators **412a** to control access to items within a row of covered compartments **408a**. The open-topped compartments **408b** may be disposed within the drawer **406b** in any arrangement. Oftentimes, open-topped compartments **408b** are used to store non-controlled substances and/or other items that do not require the additional security of a covered compartment **408a**. To provide access to an item in an open-topped compartment **408b**, the computing device **404** merely needs to unlock the relevant drawer **406b**.

Medication storage system **400** also includes a drawer **406c** that includes covered compartments **408a**. Here, columns of the covered compartments **408a** have corresponding covers **410b** and actuators **412b**. Covers **410b** and actuators **412b** operate similarly to covers **410a** and actuators **412a**, however the covers **410b** are arranged over columns of compartments **408a** and the actuators **412b** retract and extend the covers **410b** along a length of the drawer **406c**, rather than the width of the drawer **406c**. Medication storage system **400** also includes a drawer **406d** that includes only open-topped compartments **408b**. It will be appreciated that the arrangement of various drawers **406**, compartments **408**, covers **410**, and actuators **412** provided above is merely meant to serve as one example, and that numerous variations are possible.

In embodiments, once the user makes his selection of one or more items to retrieve from the medication storage system **400**, the computing device **404** identifies one or more drawers **406** and/or compartments **408** that are associated with the selected items. The computing device **404** may then send a signal to an actuator **412** associated with the relevant compartments **408**. The signal causes the relevant actuators **412** to activate to retract an attached cover **410** to expose a number of compartments **408** necessary to dispense the correct number of selected items. In some embodiments, multiple actuators **416** and covers **410** may be actuated in one session (either simultaneously or in sequence) to provide access to items in different rows and/or drawers **406**. Once the relevant covers **410** have been actuated, the relevant drawers **406** may be unlocked, allowing the user to access the drawers **406** and exposed compartments **408**. Once the user is done removing items, the user may close the drawer **406**. Once the computing device **112** detects that the drawers **104** are closed (such as by using one or more door close sensors), the computing device **404** may lock each

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drawer **406** and subsequently activate the actuators **412** to extend the cover **410** over the exposed compartments **408**. In such a manner, the medication storage system **400** may prevent the possibility of any pinch points that may be created while the covers **410** are being moved, as the covers **410** are only moved when the drawers **406** are closed and/or locked.

In some embodiments, the drawer **406** and/or cover **410** may include a number of sensors that are used to monitor inventory within the drawer **406** and/or whether a compartment **408** is full or empty. For example, as described in greater detail above, each compartment **408** of the drawer **406** may include a light, image, RF, and/or weight-based sensor that determines whether the compartment **408** includes an item. Similarly, a leading edge of each cover **410** may include a light, image, and/or RF-based sensor that is used to determine whether a given compartment is empty or full. In some embodiments, when the medication storage system **400** is idle, each cover **410** may be fully retracted to count the items within each drawer **406**. In other embodiments, a running inventory may be maintained based on a log of interactions with the medication storage system **400**. Based on knowledge of whether each compartment **408** is empty or full, the inventory of a drawer **406** may be determined. This inventory information may be tracked, oftentimes along with information about which users took the items and possibly for what purpose (task, patient, etc.) to monitor how many items are present within the drawer **406** and/or medication storage system **400** and how items have been utilized. In some embodiments, the inventory and/or usage data may be updated each time the covers **410** are extended and/or retracted and/or when each drawer **406** is opened and/or closed.

In some embodiments, the inventory may be updated periodically, such as every hour, work shift, day, etc. In other embodiments, the inventory may be tracked continuously before, during, and/or each interaction with the medication storage system **400**. In some instances, the central pharmacy **402** may initiate a remote inventory scan of one or more of the medication storage systems **400** in the facility. For example, the central pharmacy **402** may send a command to each medication storage system **400** that causes the latest inventory information from the respective medication storage system **400** to be sent to the central pharmacy. In some instances, the command may cause each medication storage system **400** to actuate sensors and/or covers **410** to determine a current inventory count, such as by fully retracting and/or extending each cover **410** to count the various items. This inventory information may be transmitted to the central pharmacy **402**. The central pharmacy may monitor the inventory of each medication storage system **400** and use the information to determine when to refill a particular item in a particular drawer **406** and/or medication storage system **400**. Additionally, such information may be useful in determining when to reorder a particular item, as the facility-wide inventory may be known based on the inventory of each medication storage system **400** and of the central pharmacy.

In some embodiments, the knowledge of the empty/full state of each compartment **408** enables the cover **410** to be actuated to expose a number of compartments **408** that allows access to the selected number of items. For example, if a drawer **406** includes ten single-item compartments **408** covered by a single cover **410**, the computing device **404** may check to see if any of the compartments **408** are empty by using one or more sensors. For example, if each of the ten compartments **408** includes an item, the computing device **404** may retract the cover **410** sufficiently far to provide

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access to the first three compartments **408**, thereby allowing the user to remove the three items as requested. In another situation, the computing device **404** may determine that the first two compartments **408** are empty and may then retract the cover **410** sufficiently far to provide access to the first five compartments **408** (including the two empty compartments **408**), thereby allowing the user to remove the three items as requested.

FIG. **15** is a flowchart illustrating a process **500** of operating a secure dispensing mechanism. The secure dispensing mechanism may be integrated into a storage unit, such as cabinet **100**, medication storage system **400**, a cart, shelving, and/or other device that may include a number of secure dispensing units (such as drawers **104**, **204**). The secure dispensing mechanism may include one or more processors or other controllers (such as computing devices **112**, **404**) that execute instructions to perform some or all of the steps of process **500**. At block **502**, an input may be received to dispense a selected number of items from a secure dispensing mechanism. The input may specify which items are to be removed and how many items are to be removed. The input may be received by a computing device, and oftentimes may be received from an authorized user. The authorized user may have previously been authenticated by providing some sort of access credentials to the computing device, such as a piece of information, an identifier from a possession-based credential, and/or a biometric credential. In some embodiments, the authenticated user may be required to enter additional information associated with a particular item, such as a patient and/or task that the item is to be used with.

After any user authentication has been performed and item selections have been received, the secure dispensing mechanism may identify one or more compartments and/or drawers within the storage unit that contain the selected items at block **504**. For example, the computing device may access a local and/or remote database of items (such as medications) and drawer/compartiment locations. In some embodiments, each drawer, cassette, and/or bin may include a computer readable identifier that informs the computing device what drawer/item arrangements are present. Once the computing device identifies a proper drawer and/or compartment that contains the selected item(s), the computing device may determine how much a cover associated with the relevant compartment needs to be retracted at block **506**.

In some embodiments, the cover may be extended to cover all compartments, including empty compartments, after each interaction. In such embodiments, the computing device may receive sensor data that is used to determine how much to retract the cover to expose the selected items at block **508**. For example, if each compartment includes a single item and there are no empty compartments, the cover may be retracted to expose a number of compartments matching the number of items selected. In other embodiments, the number of compartments may not match the number of items selected. For example, one or more of the compartments may include multiple items. In other embodiments, the secure dispensing mechanism may determine that at least one of the plurality of compartments is empty and adjust the selected number of the plurality of compartments to expose based on the determination that the at least one of the plurality of compartments is empty. For example, a row of ten compartments each includes a single item and a user wishes to access three items in the row, the secure dispensing unit may check to see if any of the compartments are empty by using one or more sensors, such as optical sensors, radio frequency (RF) sensors, load sensors, and the like. For

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example, if each of the ten compartments includes an item, the secure dispensing mechanism may retract the cover sufficiently far to provide access to the first three compartments, thereby allowing the user to remove the three items as requested. In another situation, the secure dispensing mechanism may determine that the first two compartments are empty and may then retract the cover sufficiently far to provide access to the first five compartments (including the two empty compartments), thereby allowing the user to remove the three items as requested. Once the proper retraction distance is determined, the computing device may send a command to an actuator associated with the compartment to actuate a cover to retract a leading edge of the cover to expose a selected number of compartments at block **510**.

In determining that at least one of the plurality of compartments is empty, the secure dispensing mechanism may detect an optical signal from one or more emitter devices positioned within the at least one of the plurality of compartments. In some embodiments, an optical sensor positioned on the leading edge of the cover as the cover is moved along a length of the secure dispensing mechanism to detect the optical sensor. In other embodiments, imaging sensors, weight sensors, and/or RF sensors may be integrated into the cover and/or compartments to assist the computing device in determining whether a particular bin is empty.

In other embodiments, once the cover is partially retracted, the cover will remain in place until another interaction with the associated row of compartments (such as another selection to remove an item and/or refilling the empty items), thereby keeping empty containers exposed. In such embodiments, no sensor data is needed and the cover may be actuated to expose a number of compartments matching the number of items at block **510**.

In some embodiments, multiple covers may be actuated and/or multiple rows of compartments may be exposed. In embodiments in which multiple covers (in the same row and/or different rows) are actuated, the covers may be actuated in sequence and/or simultaneously to provide access to all selected items at once. In some embodiments, actuating the cover may include rotating a sprocket gear drive that is interfaced with an edge of the cover to pull at least a portion of the cover about a portion of the sprocket gear drive.

In some embodiments, the cover and compartments may be integrated into an electronically lockable drawer of an inventory control system (such as cabinet **100** and/or medication storage system **400**). In such embodiments, the process **400** may further include unlocking the lockable drawer after the cover of the secure dispensing mechanism has been retracted at block **512**. The user then has access to the interior of the drawer and the exposed compartments. After the user has removed the desired items, the drawer is shut. Once the computing device detects that a drawer has been shut, the computing device may send a command that locks the drawer, while in other embodiments the drawer may have a self-locking mechanical lock. Once, the drawer has been locked, in some embodiments, the computing device may optionally extend the cover over each of the compartments. In some embodiments, the secure dispensing mechanism may receive an additional input to refill at least some of the compartments. The cover may be retracted to expose a number of compartments that are to be refilled, which may include all of the compartments or only a portion thereof.

In some embodiments, the process **500** may also include scanning each of the compartments while the dispensing mechanism is closed and idle to determine the contents of

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the dispensing mechanism. For example, each of the covers within a given drawer may be actuated (in sequence, in groups, or all covers simultaneously) such that the various sensors may detect the contents of each compartment for inventory purposes. In such a manner, the contents of each drawer and/or cabinet may be captured while the cabinet is not being used. In some embodiments, the results of the inventory check may be sent to a central server and/or other remote computer, which may allow for various processes, such as refilling cassettes and/or drawers and/or reordering items to be triggered. In some embodiments, such inventory procedures may be done at predetermined times and/or intervals, after a particular use of the cabinet and/or drawer, and/or based on a request from a remote computing device (such as a central server and/or pharmacy computer).

FIG. 16 is a flowchart illustrating a process 600 for using a secure dispensing mechanism. The secure dispensing mechanism may be integrated into a storage unit, such as cabinet 100, medication storage system 400, a cart, shelving, and/or other device that may include a number of secure dispensing units (such as drawers 104, 204). Process 600 may begin at block 602 by a user logging into the storage unit. For example, the user may use one or more input devices of a computing device of the storage unit to provide knowledge-based, biometric, and/or possession-based credentials as described in greater detail in relation to FIG. 1 above. Once authenticated, the user may then select use a graphical user interface (GUI) and/or other interface of the storage unit to make selections about a patient, procedure, and/or items to be removed or otherwise dispensed from the storage unit at block 604. These selections may be made using any of the input devices of the computing device, including a microphone to enable voice commands. After selections are made, the user may open one or more drawers of the storage unit that have been unlocked by the computing system at block 606. The user may then remove and/or otherwise access any items that are provided in exposed compartments within the open drawer at block 608. For example, a cover of one or more of the compartments may have been retracted by the computing device prior to the user opening the drawer. The user may then take items from the exposed compartments. In some embodiments, lights and/or other indicators may be illuminated and/or otherwise activated to guide the user to the compartment(s) containing the selected items. Once the user has all of the selected items from a given drawer, the user may close the drawer at block 610, which may then be locked by the computing device.

A computer system as illustrated in FIG. 17 may be incorporated as part of the previously described computerized devices. For example, computer system 700 can represent some of the components of computing device 112, cabinet 100, and/or other computing devices described herein. FIG. 17 provides a schematic illustration of one embodiment of a computer system 700 that can perform the methods provided by various other embodiments, as described herein. FIG. 17 is meant only to provide a generalized illustration of various components, any or all of which may be utilized as appropriate. FIG. 17, therefore, broadly illustrates how individual system elements may be implemented in a relatively separated or relatively more integrated manner. The computer system 700 is shown comprising hardware elements that can be electrically coupled via a bus 705 (or may otherwise be in communication, as appropriate). The hardware elements may include a processing unit 710, including without limitation one or more processors, such as one or more special-purpose processors (such as digital signal processing chips, graphics

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acceleration processors, and/or the like); one or more input devices 715, which can include without limitation a keyboard, a touchscreen, receiver, a motion sensor, a camera, a smartcard reader, a contactless media reader, and/or the like; and one or more output devices 720, which can include without limitation a display device, a speaker, a printer, a writing module, and/or the like.

The computer system 700 may further include (and/or be in communication with) one or more non-transitory storage devices 725, which can comprise, without limitation, local and/or network accessible storage, and/or can include, without limitation, a disk drive, a drive array, an optical storage device, a solid-state storage device such as a random access memory ("RAM") and/or a read-only memory ("ROM"), which can be programmable, flash-updateable and/or the like. Such storage devices may be configured to implement any appropriate data stores, including without limitation, various file systems, database structures, and/or the like.

The computer system 700 might also include a communication interface 730, which can include without limitation a modem, a network card (wireless or wired), an infrared communication device, a wireless communication device and/or chipset (such as a Bluetooth™ device, an 802.11 device, a Wi-Fi device, a WiMAX device, an NFC device, cellular communication facilities, etc.), and/or similar communication interfaces. The communication interface 730 may permit data to be exchanged with a network (such as the network described below, to name one example), other computer systems, and/or any other devices described herein. In many embodiments, the computer system 700 will further comprise a non-transitory working memory 735, which can include a RAM or ROM device, as described above.

The computer system 700 also can comprise software elements, shown as being currently located within the working memory 735, including an operating system 740, device drivers, executable libraries, and/or other code, such as one or more application programs 745, which may comprise computer programs provided by various embodiments, and/or may be designed to implement methods, and/or configure systems, provided by other embodiments, as described herein. Merely by way of example, one or more procedures described with respect to the method(s) discussed above might be implemented as code and/or instructions executable by a computer (and/or a processor within a computer); in an aspect, then, such special-purpose code and/or instructions can be used to configure and/or adapt a computing device to a special purpose computer that is configured to perform one or more operations in accordance with the described methods.

A set of these instructions and/or code might be stored on a computer-readable storage medium, such as the storage device(s) 725 described above. In some cases, the storage medium might be incorporated within a computer system, such as computer system 700. In other embodiments, the storage medium might be separate from a computer system (e.g., a removable medium, such as a compact disc), and/or provided in an installation package, such that the storage medium can be used to program, configure and/or adapt a special purpose computer with the instructions/code stored thereon. These instructions might take the form of executable code, which is executable by the computer system 700 and/or might take the form of source and/or installable code, which, upon compilation and/or installation on the computer system 700 (e.g., using any of a variety of available compilers, installation programs, compression/decompression utilities, etc.) then takes the form of executable code.

Substantial variations may be made in accordance with specific requirements. For example, customized hardware might also be used, and/or particular elements might be implemented in hardware, software (including portable software, such as applets, etc.), or both. Moreover, hardware and/or software components that provide certain functionality can comprise a dedicated system (having specialized components) or may be part of a more generic system. For example, a risk management engine configured to provide some or all of the features described herein relating to the risk profiling and/or distribution can comprise hardware and/or software that is specialized (e.g., an application-specific integrated circuit (ASIC), a software method, etc.) or generic (e.g., processing unit 710, applications 745, etc.) Further, connection to other computing devices such as network input/output devices may be employed.

Some embodiments may employ a computer system (such as the computer system 700) to perform methods in accordance with the disclosure. For example, some or all of the procedures of the described methods may be performed by the computer system 700 in response to processing unit 710 executing one or more sequences of one or more instructions (which might be incorporated into the operating system 740 and/or other code, such as an application program 745) contained in the working memory 735. Such instructions may be read into the working memory 735 from another computer-readable medium, such as one or more of the storage device(s) 725. Merely by way of example, execution of the sequences of instructions contained in the working memory 735 might cause the processing unit 710 to perform one or more procedures of the methods described herein.

The terms “machine-readable medium” and “computer-readable medium,” as used herein, refer to any medium that participates in providing data that causes a machine to operate in a specific fashion. In an embodiment implemented using the computer system 700, various computer-readable media might be involved in providing instructions/code to processing unit 710 for execution and/or might be used to store and/or carry such instructions/code (e.g., as signals). In many implementations, a computer-readable medium is a physical and/or tangible storage medium. Such a medium may take many forms, including but not limited to, non-volatile media, volatile media, and transmission media. Non-volatile media include, for example, optical and/or magnetic disks, such as the storage device(s) 725. Volatile media include, without limitation, dynamic memory, such as the working memory 735. Transmission media include, without limitation, coaxial cables, copper wire, and fiber optics, including the wires that comprise the bus 705, as well as the various components of the communication interface 730 (and/or the media by which the communication interface 730 provides communication with other devices). Hence, transmission media can also take the form of waves (including without limitation radio, acoustic and/or light waves, such as those generated during radio-wave and infrared data communications).

Common forms of physical and/or tangible computer-readable media include, for example, a magnetic medium, optical medium, or any other physical medium with patterns of holes, a RAM, a PROM, EPROM, a FLASH-EPROM, any other memory chip or cartridge, a carrier wave as described hereinafter, or any other medium from which a computer can read instructions and/or code.

The communication interface 730 (and/or components thereof) generally will receive the signals, and the bus 705 then might carry the signals (and/or the data, instructions, etc. carried by the signals) to the working memory 735, from

which the processor(s) 710 retrieves and executes the instructions. The instructions received by the working memory 735 may optionally be stored on a non-transitory storage device 725 either before or after execution by the processing unit 710.

The methods, systems, and devices discussed above are examples. Some embodiments were described as processes depicted as flow diagrams or block diagrams. Although each may describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be rearranged. A process may have additional steps not included in the figure. Furthermore, embodiments of the methods may be implemented by hardware, software, firmware, middleware, microcode, hardware description languages, or any combination thereof. When implemented in software, firmware, middleware, or microcode, the program code or code segments to perform the associated tasks may be stored in a computer-readable medium such as a storage medium. Processors may perform the associated tasks.

It should be noted that the systems and devices discussed above are intended merely to be examples. It must be stressed that various embodiments may omit, substitute, or add various procedures or components as appropriate. Also, features described with respect to certain embodiments may be combined in various other embodiments. Different aspects and elements of the embodiments may be combined in a similar manner. Also, it should be emphasized that technology evolves and, thus, many of the elements are examples and should not be interpreted to limit the scope of the invention.

Specific details are given in the description to provide a thorough understanding of the embodiments. However, it will be understood by one of ordinary skill in the art that the embodiments may be practiced without these specific details. For example, well-known structures and techniques have been shown without unnecessary detail in order to avoid obscuring the embodiments. This description provides example embodiments only, and is not intended to limit the scope, applicability, or configuration of the invention. Rather, the preceding description of the embodiments will provide those skilled in the art with an enabling description for implementing embodiments of the invention. Various changes may be made in the function and arrangement of elements without departing from the spirit and scope of the invention.

The methods, systems, devices, graphs, and tables discussed above are examples. Various configurations may omit, substitute, or add various procedures or components as appropriate. For instance, in alternative configurations, the methods may be performed in an order different from that described, and/or various stages may be added, omitted, and/or combined. Also, features described with respect to certain configurations may be combined in various other configurations. Different aspects and elements of the configurations may be combined in a similar manner. Also, technology evolves and, thus, many of the elements are examples and do not limit the scope of the disclosure or claims. Additionally, the techniques discussed herein may provide differing results with different types of context awareness classifiers.

While illustrative and presently preferred embodiments of the disclosed systems, methods, and machine-readable media have been described in detail herein, it is to be understood that the inventive concepts may be otherwise variously embodied and employed, and that the appended

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claims are intended to be construed to include such variations, except as limited by the prior art.

Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly or conventionally understood. As used herein, the articles “a” 5 and “an” refer to one or to more than one (i.e., to at least one) of the grammatical object of the article. By way of example, “an element” means one element or more than one element. “About” and/or “approximately” as used herein when referring to a measurable value such as an amount, a temporal 10 duration, and the like, encompasses variations of $\pm 20\%$ or $\pm 10\%$, $\pm 5\%$, or $\pm 0.1\%$ from the specified value, as such variations are appropriate to in the context of the systems, devices, circuits, methods, and other implementations described herein. “Substantially” as used herein when referring to a measurable value such as an amount, a temporal 15 duration, a physical attribute (such as frequency), and the like, also encompasses variations of $\pm 20\%$ or $\pm 10\%$, $\pm 5\%$, or $\pm 0.1\%$ from the specified value, as such variations are appropriate to in the context of the systems, devices, circuits, methods, and other implementations described herein. As used herein, including in the claims, “and” as used in a list of items prefaced by “at least one of” or “one or more of” indicates that any combination of the listed items may be used. For example, a list of “at least one of A, B, and C” 25 includes any of the combinations A or B or C or AB or AC or BC and/or ABC (i.e., A and B and C). Furthermore, to the extent more than one occurrence or use of the items A, B, or C is possible, multiple uses of A, B, and/or C may form part of the contemplated combinations. For example, a list of “at least one of A, B, and C” may also include AA, AAB, AAA, BB, etc.

Having described several embodiments, it will be recognized by those of skill in the art that various modifications, alternative constructions, and equivalents may be used without departing from the spirit of the invention. For example, the above elements may merely be a component of a larger system, wherein other rules may take precedence over or otherwise modify the application of the invention. Also, a number of steps may be undertaken before, during, or after 40 the above elements are considered. Accordingly, the above description should not be taken as limiting the scope of the invention.

Also, the words “comprise”, “comprising”, “contains”, “containing”, “include”, “including”, and “includes”, when used in this specification and in the following claims, are intended to specify the presence of stated features, integers, components, or steps, but they do not preclude the presence or addition of one or more other features, integers, components, steps, acts, or groups. 50

What is claimed is:

1. A dispensing unit, comprising:

- a housing comprising a first end and a second end; and
- a plurality of storage assemblies, wherein the plurality of storage assemblies are arranged in parallel with one another within the housing, each storage assembly comprising:
 - a plurality of compartments arranged linearly along an axis of the housing, each of the compartments being 60 configured to store an item;
 - a cover coupled with the housing and extending between the first end and the second end such that the cover is positioned over the plurality of compartments;
 - a plurality of RF sensors that are configured to determine whether one or more of the plurality of com-

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partments are empty, wherein each of the plurality of RF sensors is disposed in one of the plurality of compartments; and

an actuator coupled with the cover, the actuator being configured to retract the cover to draw a distal end of the cover from the first end toward the second end to expose a number of the plurality of compartments to expose a selected number of items stored in the plurality of compartments based on a reading from the plurality of RF sensors.

2. The dispensing unit of claim 1, wherein:

each of the plurality of compartments comprises a base and one or more walls; and

the base and the one or more walls comprise RF shielding.

3. The dispensing unit of claim 1, wherein:

the actuator comprises an encoder that determines how far the cover is retracted to expose the selected number of the plurality of compartments.

4. The dispensing unit of claim 1, wherein:

the plurality of RF sensors determine whether one or more of the plurality of compartments are empty based on whether an RF tag attached to an item is read within each of the one or more of the plurality of compartments.

5. The dispensing unit of claim 1, wherein:

the plurality of RF sensors are further configured to determine an inventory of a respective one of the plurality of storage assemblies.

6. A dispensing unit, comprising:

- a housing comprising a first end and a second end;
- a plurality of compartments arranged linearly along an axis of the housing, each of the compartments being configured to store an item;

- a cover coupled with the housing and extending between the first end and the second end such that the cover is positioned over the plurality of compartments;

- an actuator coupled with the cover;

- at least one RF sensor;

- one or more processors; and

- a memory having instructions stored thereon that, when executed by the one or more processors, cause the one or more processors to:

- determine whether one or more of the plurality of compartments are empty using the at least one RF sensor; and

- cause the actuator to retract the cover to draw a leading edge of the cover from the first end toward the second end to expose a selected number of items stored in the plurality of compartments based on determining whether one or more of the plurality of compartments are empty; and

- determine an inventory of the dispensing unit using the at least one RF sensor.

7. The dispensing unit of claim 6, wherein:

the dispensing unit is integrated into an electronically lockable drawer of a medication storage system.

8. The dispensing unit of claim 6, wherein:

- the at least one RF sensor comprises an antenna that is disposed on a leading edge of the cover; and

- the one or more processors determine whether one or more of the plurality of compartments are empty as the leading edge of the cover is moved along a length of the housing.

9. The dispensing unit of claim 6, wherein:

- each of the plurality of compartments comprises a base and one or more walls; and

- the base and the one or more walls comprise RF shielding.

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10. The dispensing unit of claim 6, wherein:
 the at least one RF sensor comprises a plurality of RF
 sensors; and
 each of the plurality of RF sensors is disposed in one of
 the plurality of compartments. 5
11. The dispensing unit of claim 6, wherein:
 the one or more processors determine whether one or
 more of the plurality of compartments are empty based
 on whether the at least one RF sensor reads an RF tag
 attached to an item within each of the one or more of 10
 the plurality of compartments.
12. A method of operating a dispensing mechanism,
 comprising:
 receiving an input to dispense a selected number of items 15
 from a dispensing mechanism;
 actuating a cover of the dispensing mechanism to retract
 a leading edge of the cover to expose a selected number
 of a plurality of compartments of the dispensing
 mechanism based on the selected number of items, 20
 wherein the plurality of compartments are arranged
 linearly along an axis of the dispensing mechanism,
 each of the compartments being configured to store an
 item; and
 determining that at least one of the plurality of compart- 25
 ments is empty using at least one RF sensor of the
 dispensing mechanism, wherein:

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- the at least one RF sensor comprises an antenna that is
 disposed on a leading edge of the cover; and
 determining that at least one of the plurality of compart-
 ments is empty comprises failing to read an RF
 tag of an item within the at least one of the plurality
 of compartments using the antenna as the leading
 edge of the cover is moved along the axis of the
 dispensing mechanism.
13. The method of operating a dispensing mechanism of
 claim 12, further comprising:
 adjusting the selected number of the plurality of compart-
 ments to expose based on the determination that the at
 least one of the plurality of compartments is empty.
14. The method of operating a dispensing mechanism of
 claim 12, further comprising:
 receiving an additional input to refill the plurality of 15
 compartments; and
 actuating the cover of the dispensing mechanism to retract
 the leading edge of the cover to expose each of the
 plurality of compartments in response to the additional
 input.
15. The method of operating a dispensing mechanism of
 claim 12, wherein:
 each of the plurality of compartments comprises a base
 and one or more walls; and the base and the one or more
 walls comprise RF shielding.

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